

The Health Costs of Firms

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Abstract

This paper estimates how much firms differ in their contribution to otherwise similar workers' healthcare expenditure. Using linked Dutch employer–employee data, I exploit worker moves across firms to separate persistent worker and firm differences. Moving the same worker to a firm one standard deviation higher in the firm-effect distribution raises expected annual expenditure by 17.8%. The implied top–bottom quintile gap is comparable to gradients by income, wealth, and education, and over half of the dispersion lies within narrowly defined industries. Supporting a health-risk interpretation, workers who move to firms with higher coworker expenditure increase use of pain, anti-inflammatory, and muscle relaxant medications, and exposure to high-effect firms predicts disability and mortality, including among workers displaced by closures. High-effect firms pay higher wage premia, but mainly between rather than within industries. Because basic healthcare is collectively financed, the results point to a fiscal externality: firms generate healthcare expenditure they do not fully bear.

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I Introduction

Firms affect workers' health through physical demands, psychosocial stressors, and management practices, and a growing body of work links workplace conditions to healthcare use, injury, disability, and mortality ([Hummels et al., 2025](#); [Sullivan and Von Wachter, 2009](#); [Johnson, 2020](#); [Jones et al., 2019](#); [Ahammer et al., 2026](#)).¹ This paper shows that the firm a worker is employed at is a quantitatively important determinant of his healthcare expenditure and subsequent health, and that the resulting costs are neither fully priced in wages nor borne by the firm: they spill over to the publicly financed healthcare system. Moving the same worker across the distribution of firms shifts his expected healthcare expenditure by as much as moving him (descriptively) across the distribution of income or education in the sample; more than half of this dispersion lies among firms in the same narrowly defined industry; within industries, firms with higher health-cost effects pay little or no additional wage premium; and in a publicly financed system, the medical expenditure associated with workplace environments is borne by the collective insurer rather than by the firm.

Using linked employer–employee administrative data from the Netherlands, this paper delivers the first estimates of the dispersion of firm effects on annual healthcare expenditure: how much expenditure varies across firms for otherwise similar workers.² The data link every full-time male employee aged 25 to 65 to his employer and to his basic-insurance claims, covering 17 million worker-years, 1.2 million movers, and 133,000 firms of all sizes connected by worker mobility. I leverage moves across firms to separate persistent worker differences from persistent firm differences, thereby recovering the distribution of firm effects in

¹Firms can also affect healthcare access through employer-sponsored insurance and provider networks ([Tilipman, 2022](#); [Ho and Lee, 2019](#)). As discussed below, this channel is largely absent in the Dutch setting.

²This object differs from the coefficient of mover-design event studies ([Finkelstein et al., 2016](#); [Ahammer et al., 2026](#)), which measures how much of the observed gap between origin and destination firms passes through to movers. In a companion paper ([Moghani, 2026](#)), I show that this coefficient maps into a cross-sectional share only under a testable restriction on mobility patterns—movers must traverse firm effects and worker composition in the same proportions as the cross-section—and that even then it recovers a share of the dispersion in firm averages, not the contribution of firms to expenditure differences across individual workers, which is the object estimated here. Appendix [A3](#) summarises the result and the division of labour between the two designs in this paper.

healthcare expenditure.³ The firm effect answers a simple question: by how much would the same worker's expected healthcare expenditure change if he worked at a different firm? The Dutch setting is well suited to this question because basic health insurance is individual, universal, and portable across jobs rather than tied to employers, and primary care is free at the point of use. Cross-firm differences in expenditure are therefore less likely to reflect employer-driven differences in insurance coverage or cost sharing.

I document substantial variation across firms in healthcare expenditure among otherwise similar workers. Moving to a firm one standard deviation higher in the firm-effect distribution increases expected annual healthcare expenditure by 17.8%. In the study sample, the implied top–bottom quintile gap in firm effects is comparable in magnitude to gradients in healthcare expenditure by income, wealth, and education. Among firms that can be linked to the business registry, more than 50% of the dispersion remains within narrowly defined industrial codes, suggesting that the variation is not simply between firms' activities but also reflects differences across firms operating in similar lines of business.⁴

Supporting a health-risk interpretation, moves to firms with higher average coworker expenditure are followed by increases in pain, anti-inflammatory, and muscle relaxant medications, with little corresponding change in categories more closely tied to screening or diagnosis, a pattern suggestive of workplace-related strain or injury. In the long run, workers with greater exposure to high-effect firms are more likely to enter disability and face higher mortality, suggesting that cross-firm variation in healthcare expenditure is linked to persistent differences in health risk. The same relationship holds among workers displaced by firm closures, for whom the separation is involuntary: reemployment at a destination firm one standard deviation higher in the firm-effect distribution is associated with a 0.22 percentage point higher probability of disability in 2019 and a 0.08 percentage point higher probability of death, roughly 6% of the respective sample means. The

³Formally, I estimate a leave-one-out bias-corrected Abowd–Kramarz–Margolis (AKM) worker–firm two-way fixed effects model of log annual healthcare expenditure for full-time male workers from 2009–2016 (Abowd et al., 1999; Kline et al., 2020). I focus on men because women's job-to-job moves are closely tied to fertility decisions (Figure A7), which can violate the mobility assumptions behind the decomposition. The results for the female sample are reported in Appendix A8 and are similar to those for the male sample.

⁴I use detailed 898 Dutch standard industrial classification (SBI) codes, which are comparable in spirit to SIC or NAICS industry classifications.

firm gradient is also visible in GP expenditure, which is free at the point of use, and among high-spending workers who face near-zero marginal prices, suggesting that these differences are not solely driven by financial incentives.

Two features of the estimation matter for interpretation. Variance components computed from noisy firm-effect estimates overstate dispersion when mobility between firms is limited; I therefore use the leave-one-out correction of [Kline et al. \(2020\)](#), and the covariance of firm effects estimated in two independent halves of the workforce (0.026) is close to the corrected variance (0.027). For the disability and mortality analyses, including the closure analysis, I use a sample-split procedure following [Kline \(2025\)](#): firm effects estimated from one group of workers are used to predict outcomes in the other, so that a worker's own healthcare history does not enter the firm-effect measure assigned to him.

The main estimates are robust when restricting the sample to younger workers, for whom predictable health declines are less likely to drive job switches. Event-study evidence around job-to-job moves also shows no differential pre-trends in workers' healthcare expenditure prior to the move, and the pre-move expenditure of displaced workers is not higher at destination firms with higher effects; together, these patterns are consistent with mobility not being driven by emerging health problems in aggregate. The log-additive specification is supported by a saturated model with worker-by-firm effects, which raises the adjusted R^2 only from 0.516 to 0.532 despite 56% more parameters, by mean residuals close to zero across deciles of worker and firm effects, and by the approximately linear relationship between movers' own expenditure changes and coworker expenditure gaps. The estimated dispersion is essentially unchanged when controlling for annual pay and changes little with residential postcode fixed effects and workplace-area measures of healthcare access (0.025); firm effects estimated separately by collar type, wage, baseline expenditure, and age rank firms similarly; the corrected dispersion is stable on a highly connected subset of firms; and firm effects on future expenditure attenuate with the horizon, consistent with contemporaneous workplace effects.

This paper contributes to the literature on firm heterogeneity and inequality by adding healthcare expenditure and health to the set of worker outcomes shaped by firms. Existing work shows that firms pay different wage premia to otherwise similar workers and account for a substantial share of earnings

inequality (Kline, 2024; Card et al., 2013, 2018, 2016), differ in amenities and working conditions (Sockin, 2022; Lamadon et al., 2022), and contribute to inequality through differential treatment by gender, origin, and other characteristics (Kline et al., 2022). The evidence on whether non-wage attributes offset or reinforce wage differences is mixed (Sockin, 2022; Lamadon et al., 2022; Sorkin, 2018). Surprisingly little is known, however, about whether otherwise similar workers incur systematically different healthcare expenditure depending on the firm that employs them. Bíró and Elek (2025) show, in Hungarian data, that moving to a higher-quality firm—measured by its wage premium—raises the detection of chronic conditions through more physician visits and cardiovascular medication, a care-seeking channel plausibly operating through occupational health check-ups. The firm effects estimated here are a different object, the dispersion of firms’ contributions to expenditure rather than the effect of a particular firm attribute, and the medication evidence points to health risk rather than detection: cardiometabolic and cancer-related medication use does not respond to moves, whereas pain, anti-inflammatory, and muscle relaxant medication use does. This paper provides the first estimates of that dispersion, documents that it carries longer-run health implications, and shows that higher wage premia are associated with higher health-cost effects, consistent with compensating differentials (Rosen, 1974; Marin and Psacharopoulos, 1982; Arnould and Nichols, 1983; Viscusi and Moore, 1987; Mas and Pallais, 2017). This association arises between industries, however: among firms within the same detailed industry, where more than half of the dispersion in firm effects lies, it is close to zero, consistent with compensation for risks that are visible at the industry level rather than for firm-specific risk.

The findings also connect the firm-heterogeneity literature to the health-inequality literature, where income, education, occupation, and place are documented as central sources of inequality (Cutler et al., 2006; Lleras-Muney et al., 2025; O’Donnell et al., 2015; Ravesteijn et al., 2018; Finkelstein et al., 2021, 2016). While earlier work identifies causal effects of specific workplace conditions on average employee health (e.g., Hummels et al., 2025; Lee and Taylor, 2019), this paper estimates the contribution of the firm environment to healthcare-expenditure disparities between otherwise similar workers, with evidence that these differences are related to broader health risks. In contemporaneous and independent work,

[Ahammer et al. \(2026\)](#) use Austrian linked data and a mover event-study design to show that about a quarter of the observed expenditure gap between origin and destination firms is transmitted to movers, that accidents and injuries account for much of the response, and that firms with higher estimated contributions have higher worker mortality and pay higher wage premia. This paper estimates a different object—the dispersion of firms’ own contributions for a given worker—on the national workforce, predicts each worker’s disability and mortality with firm effects estimated on other workers, and adds firm closures, the within-industry structure of the dispersion, and the between- and within-industry pattern of wage premia; the two papers’ findings on accidents, injury-related medication, and mortality reinforce each other.

The results point to a fiscal externality in collectively financed healthcare systems such as the Dutch one. Because health insurance contributions are not tied to realised medical expenditure at the firm level, firms do not bear the marginal cost of the healthcare expenditure generated by their workforce, which weakens incentives to reduce workplace health risks even under stringent safety regulation and experience rating in sickness and disability insurance. The fact that more than half of the dispersion remains within narrowly defined industrial codes suggests that this problem is not only about industry composition but also about firm-specific practices and organisational design. While wages may partly compensate workers for the disutility of health risks—and within industries appear not to—they do not reimburse the public financing system for the resulting medical costs. This creates an unpriced fiscal externality and points to the value of policies that better align firms’ incentives with the healthcare expenditure they generate.

II Background

This section summarizes the institutional features of the Dutch setting that matter for interpreting the estimates. Appendix [A1](#) provides additional detail on the healthcare system, occupational-health obligations, sickness provisions, and the channels through which firms may affect employees’ health and healthcare expenditures.

The Netherlands has a system of universal basic health insurance provided by private insurers under tight public regulation. Basic insurance is individual and

portable across jobs: coverage does not change when workers change employers, and the statutory content of the basic package is the same across insurers. Cost sharing in the basic scheme is largely standardized through annual deductibles, but GP care is fully covered and exempt from the deductible. Access to care is organized around the general practitioner, who serves as the first point of contact and gatekeeper for most non-urgent specialist and hospital treatment ([Currie and Zwiers, 2025](#)). For this paper, the key implication is that cross-firm differences in observed universal basic-package expenditure arise in a setting with limited employer-driven variation in insurance generosity, provider networks, or statutory cost sharing.

At the same time, Dutch employers face extensive obligations in occupational health and sickness management. Under the Working Conditions Act (*Arbowet*), firms must monitor workplace risks, contract occupational-health services or experts, provide access to a company doctor, and continue wage payments for up to two years during illness. These obligations are reinforced by experience rating in disability-related insurance. Yet day-to-day healthcare still runs primarily through the ordinary GP-based system, and occupational doctors provide work-related advice and reintegration support rather than medical treatment. The Dutch institutional setting therefore combines substantial employer responsibility for prevention and sickness management with limited employer control over the content of ordinary medical care.

Firm effects on basic healthcare expenditure may reflect two broad forces: differences in care use conditional on health and differences in the health consequences of the workplace itself.⁵ The Dutch institutional setting narrows several care-use channels that are central in other settings. Basic insurance is universal and portable across jobs; employers do not determine statutory coverage, provider networks, or statutory cost sharing; GP care is fully covered; and out-of-pocket exposure is limited by the annual deductible. Existing evidence also finds limited income effects on healthcare expenditure ([Miller et al., 2024](#); [Cesarini et al., 2016](#)). Time barriers also appear modest. Collective agreements and labour law provide paid time off for sickness and medical visits, employers must continue wage payments during illness for up to two years, and reintegration is formally supervised by

⁵Appendix [A4](#) illustrates this distinction in a Grossman-style framework ([Grossman, 1972b](#)), in which the workplace enters through a health-depreciation rate and a cost of seeking care.

occupational physicians. Consistent with this, self-reported unmet medical need in the Netherlands is among the lowest in the EU and shows little income gradient (e.g. [OECD and European Observatory on Health Systems and Policies, 2019, 2021](#)); lack of time is cited by less than 1% of respondents ([Eurostat, 2023](#)). These features make employer-specific insurance design, statutory cost sharing, and broad financial access less likely to be the main sources of the estimated firm effects. They do not, however, rule out care-use explanations more generally, such as differences in screening, or workplace norms around seeking care. Nor do they identify a specific workplace health mechanism. The firm effects estimated may combine physical risks, psychosocial strain, work organisation, safety practices, peer norms, and prevention or reintegration policies (e.g. [Hamermesh, 1999](#); [Lee and Taylor, 2019](#); [Johnson, 2020](#); [Johansson et al., 2023](#); [Bhattacharya, 2014](#); [van Ours, 2019](#); [Hummels et al., 2025](#); [Blackburn et al., 2023](#); [Jolivet and Postel-Vinay, 2025](#); [Collaborators et al., 2018](#); [Loomis et al., 2018](#); [Tran et al., 2022](#); [Gan et al., 2011](#); [Royer et al., 2015](#); [Jones et al., 2019](#); [Simonsen and Skipper, 2025](#)).

III Dataset Description, Sample Selection, and Descriptive Statistics

I use Dutch administrative employer–employee data covering all male employees living in the Netherlands.⁶ The data contain information on employer–employee matches, demographics, income, contract start and end dates, and firms’ Dutch standard industrial classification (SBI) codes.⁷ I also observe workplace postal codes, individuals’ residential locations over time, highest completed education, and annual total and component-level basic healthcare expenditure (general practitioner, medication, specialized mental healthcare, and hospital care).⁸

⁶The corresponding results for female employees are reported in Appendix A8. For women, the patterns in Figure A7 indicate that firm changes are endogenous to childbearing decisions. Given the direct and indirect healthcare expenditure of childbearing ([Ahammer et al., 2023](#); [García-Gómez et al., 2026](#)), estimates based on firm moves should therefore be interpreted with caution for female workers.

⁷[Standaard Bedrijfsindeling \(SBI\)](#) is the Dutch standard industrial classification used by Statistics Netherlands and the Chamber of Commerce to assign each firm an industrial code. The hierarchy spans five levels, with up to five digits identifying increasingly specific industries (e.g. code 01131 “Growing of vegetables in open fields” versus code 01132 “Greenhouse growing of vegetables”). The analysis data contain 898 unique SBI codes.

⁸The baseline data do not contain occupation information. In Section V, I examine heterogeneity by education-proxied collar type and show that the estimated firm effects are largely shared across

Employment information is available annually from 2009 to 2016, while health-care expenditure is observed annually from 2009 to 2019. I focus on male employees aged 25–65 who, in each year, work at least 11 months in jobs with a contract of at least 0.9 full-time equivalent.^{9,10} When an individual is employed by multiple firms within the same year, I assign the worker to the firm where he spent the most months, since healthcare expenditure is observed annually.¹¹ These restrictions reduce reliance on functional-form assumptions relating work intensity to health-care expenditure and simplify the assignment of workers to firms. A consequence is that workers who move late in the year remain assigned to their origin firm for that year.

For these full-time workers, I observe background characteristics such as birth year, migration background, and—for most individuals—education.¹² The data further include annual salary, employer identifier, employer sector, and firm size.

For each worker, I observe total and category-specific basic healthcare expenditure from 2009 to 2019. I observe all claims billed to the basic insurance package.¹³ Total expenditure combines insurer reimbursements and individuals’ deductible payments, which are not separately identified in the data. The categories are: (1) primary care (general practitioner services); (2) medication; (3) specialized mental healthcare outside primary care (psychology and psychiatry); and (4) hospital care, covering specialist treatments and inpatient admissions.¹⁴ Employees in the defence sector (SBI code 8422) obtain healthcare through the Ministry of Defence,

groups. Moreover, I document substantial firm-effect dispersion within narrowly defined SBI industrial codes, suggesting that the results are not driven solely by broad industry or occupation groups.

⁹Full-time employment in the Netherlands ranges from 38 to 40 hours per week.

¹⁰Requiring at least 0.9 FTE excludes part-time workers, who constitute less than 20% of employed men between ages 25 and 65 ((CBS), 2025), and removes years in which employees shift into part-time work or leave employment. Sick employees continue to receive employer-paid wages for up to two years before entering sickness or disability schemes, so these periods remain recorded as employment, which is important for observing healthcare expenditure during sickness.

¹¹The 0.9 FTE requirement effectively rules out simultaneous substantial contracts.

¹²Education information is complete for cohorts born after 1990. For older cohorts, education is observed only for those who interacted with administrative processes requiring educational documentation. Individuals with missing education are treated as a separate category. Results are robust to excluding these individuals.

¹³Care outside the basic package is therefore not observed. This is most relevant for short-term physiotherapy, which is often financed out of pocket or through supplementary insurance.

¹⁴Together, these categories represent about 90% of total basic healthcare expenditure; remaining expenditure includes categories such as physiotherapy and patient transport.

which operates a parallel system not captured in these data. I therefore exclude defence-sector firms and their employees, who account for about 1.5% of the sample.

The sample is restricted to firms connected through worker mobility, which is required for estimating the AKM model because it relies on an employer network linked by movers ([Abowd et al., 1999](#)). To address limited-mobility bias, I apply the leave-one-out correction proposed by [Kline et al. \(2020\)](#), which removes each employer–employee match iteratively during estimation. This procedure requires that the network remains connected after excluding any given match. As a result, the sample is further restricted to ensure connectivity under leave-one-out removal. Summary statistics for the main sample, the connected sample, and the movers sample appear in [Table 1](#).

[Table 1](#) shows that restricting the data to the connected set leaves the composition of the sample essentially unchanged. Relative to the initial full-time sample, workers in connected firms have very similar age, migration background, education, birth rates, and healthcare expenditure, with only small increases in average age (44.2 vs. 44.0), wages, firm size, and total healthcare expenditure. By contrast, within the connected set, movers differ systematically from stayers. Movers are on average about three years younger (42.6 vs. 45.6), somewhat better educated (higher shares with upper secondary, bachelor, and master degrees and fewer missing-education records), and earn slightly higher hourly wages. They are also more likely to experience a birth in a given year (5% vs. 3%) and have marginally fewer children on average, consistent with mobility being concentrated among younger workers. Movers tend to work in slightly smaller firms. Finally, movers have lower total healthcare expenditure, hospital expenditure, and specialized mental-health expenditure than stayers, consistent with their younger age and suggesting that movers are somewhat positively selected on health. The AKM framework assumes that firm effects are common across subgroups, so that effects identified from movers also apply to stayers and to workers with different observable characteristics; this restriction is not directly testable for stayers, who never switch firms. In [Section V](#), I examine robustness of findings across subsamples defined by age, income, and baseline healthcare expenditure and show that the estimated firm effects and their dispersion are similar. In [Section IV](#), I also provide additional

empirical support for the common-firm-effect assumption.

	Initial Sample (Full-time Employees) N = 18,705,218	Connected Set (Connected Firms) N = 16,976,071	Stayers (Connected Firms) N = 9,236,759	Movers (Connected Firms) N = 7,739,312
Variable	Mean (SD)	Mean (SD)	Mean (SD)	Mean (SD)
Age	43.99 (10.48)	44.24 (10.39)	45.60 (10.51)	42.62 (10.01)
Dutch background (dummy)	0.83 (0.38)	0.83 (0.37)	0.83 (0.37)	0.83 (0.37)
First-generation migrant (dummy)	0.10 (0.30)	0.10 (0.30)	0.10 (0.31)	0.09 (0.29)
Second-generation migrant (dummy)	0.07 (0.25)	0.07 (0.25)	0.06 (0.24)	0.07 (0.26)
Education: Elementary (dummy)	0.02 (0.16)	0.02 (0.15)	0.02 (0.15)	0.02 (0.15)
Education: Lower sec./MBO1 (dummy)	0.06 (0.24)	0.06 (0.24)	0.06 (0.24)	0.06 (0.24)
Education: Upper sec./MBO (dummy)	0.23 (0.42)	0.23 (0.42)	0.20 (0.40)	0.26 (0.44)
Education: Bachelor (HBO/WO) (dummy)	0.17 (0.37)	0.17 (0.38)	0.15 (0.36)	0.20 (0.40)
Education: Master/Doctorate (dummy)	0.11 (0.31)	0.11 (0.31)	0.10 (0.29)	0.12 (0.33)
Education: Missing (dummy)	0.41 (0.49)	0.41 (0.49)	0.46 (0.50)	0.34 (0.47)
Total # children	1.38 (1.23)	1.40 (1.23)	1.44 (1.24)	1.35 (1.21)
New child (dummy)	0.04 (0.20)	0.04 (0.20)	0.03 (0.18)	0.05 (0.22)
Gross payment of contract (€)	49,616 (46,120)	50,545 (46,821)	50,510 (47,553)	50,586 (45,932)
Mover (dummy)	0.44 (0.50)	0.46 (0.50)	0.00 (0.00)	1.00 (0.00)
Full-time factor (FTE)	0.99 (0.02)	0.99 (0.02)	1.00 (0.02)	0.99 (0.02)
Estimated hourly wage (€) ^a	24.32 (22.55)	24.76 (22.91)	24.48 (23.00)	25.09 (22.79)
Business size (employees)	692.46 (800.51)	741.76 (806.30)	763.30 (804.44)	715.98 (807.77)
Total healthcare expenditure (€)	1,143 (4,800)	1,149 (4,780)	1,260 (5,263)	1,015 (4,127)
Mental health (specialist) (€)	84.46 (1,434)	83.22 (1,389)	90.03 (1,573)	75.10 (1,130)
Hospital expenditure (€)	687.31 (4,040)	691.63 (4,036)	767.33 (4,451)	601.28 (3,474)
GP expenditure (€)	109.70 (69.00)	110.09 (68.58)	111.42 (70.71)	108.51 (65.92)
Number of Observations	18,705,218	16,976,071	9,236,759	7,739,312

^a Hourly wage computed from gross contract payment assuming a full FTE works 52 weeks × 40 hours per week.

Notes: Universe of male employees aged 25–65 observed between 2009 and 2016. The sample is restricted to full-time jobs with FTE ≥ 0.9 and at least 11 months employed in a given year; if an individual is employed by multiple firms within the same year, the employer with the largest number of months is used. “Connected set” refers to employees in firms that belong to the leave-one-out connected component of the employer network, i.e. the set of firms that remain connected through worker moves when any single worker–firm match is removed. “Stayers” are worker–year observations for individuals who remain with the same employer throughout the 2009–2016 period; “Movers” are worker–year observations for individuals who change employer at least once over this period. Sectors are defined on the basis of the first two digits of the Dutch Standaard Bedrijfsindeling (SBI) classification, grouped into 17 broad sectors. Business size is measured as the midpoint of the categorical firm–size intervals reported in the data (interpreted as the approximate number of employees at the firm in the given year). Healthcare variables refer to annual expenditure under the basic health insurance package (insurer payments plus deductible). “Total healthcare expenditure” is the sum of all basic-package claims; “Mental health (specialist)” includes specialized mental healthcare outside primary care; “Hospital expenditure” cover specialist and inpatient hospital care; and “GP expenditure” refer to primary care delivered by general practitioners. Employees in the defence sector (SBI 8422), who obtain healthcare through a separate system, are excluded from all samples.

Table 1: Summary Statistics

For the main analysis, I model annual healthcare expenditure in logs. Under a multiplicative structure of health investments, such as Cobb–Douglas, taking

logs yields additive individual and firm components.¹⁵ This property is important for interpreting firm effects as “shared” across subgroups—such as blue- and white-collar workers, high- and low-income workers, or high- and low-spending individuals—because the same firm component on the log scale implies proportionally similar effects while preserving larger level effects for workers with higher individual expenditure.¹⁶¹⁷

A common concern with log transformations is the treatment of zero outcomes, since estimates can then depend on the measurement unit and on the mix of extensive- and intensive-margin responses (Chen and Roth, 2024; Mullahy and Norton, 2024). In my data, however, the share of observations with zero total healthcare expenditure is very small (Figure A1), so this issue is limited in practice. Using contemporaneous annual healthcare expenditure is also appropriate in this setting. Dutch employers must continue wage payments for up to two years during sickness, so periods of poor health remain attached to the firm where they arise. Moreover, Table A16 shows that firm effects on future healthcare expenditure are noticeably smaller than firm effects on contemporaneous expenditure. This supports using contemporaneous healthcare expenditure as the baseline outcome and interpreting the estimated firm effects as contemporaneous workplace-related contributions to healthcare expenditure.

Firms differ in the average basic healthcare expenditure of their workers, both within and across sectors (Figure 1). These raw differences, however, cannot be interpreted as firm effects. Workers and firms sort: lower-skilled and otherwise disadvantaged workers are more likely to be employed in low-wage, low-amenity firms (Card et al., 2013; Sockin, 2022), and health is strongly graded by socioeconomic status (O’Donnell et al., 2015; Danesh et al., 2024). As a result, firms employing workers with poorer underlying health or higher healthcare needs will mechanically display higher average healthcare expenditure even if the workplace itself has no effect on health. I therefore use within-worker changes in healthcare

¹⁵Appendix A4 shows, in a Grossman-style framework and up to standard approximations, that multiplicative individual and firm components in health depreciation and care-seeking costs deliver a log-additive decomposition into individual, firm, and age terms.

¹⁶Section V shows that firm effects and their variance are broadly similar across these subgroups, consistent with such a shared firm component.

¹⁷Additionally, the KSS correction is derived for OLS estimators in linear models; this motivates the use of a linear AKM specification in logs rather than nonlinear alternatives.

expenditure around firm moves to separate persistent worker components from firm components.

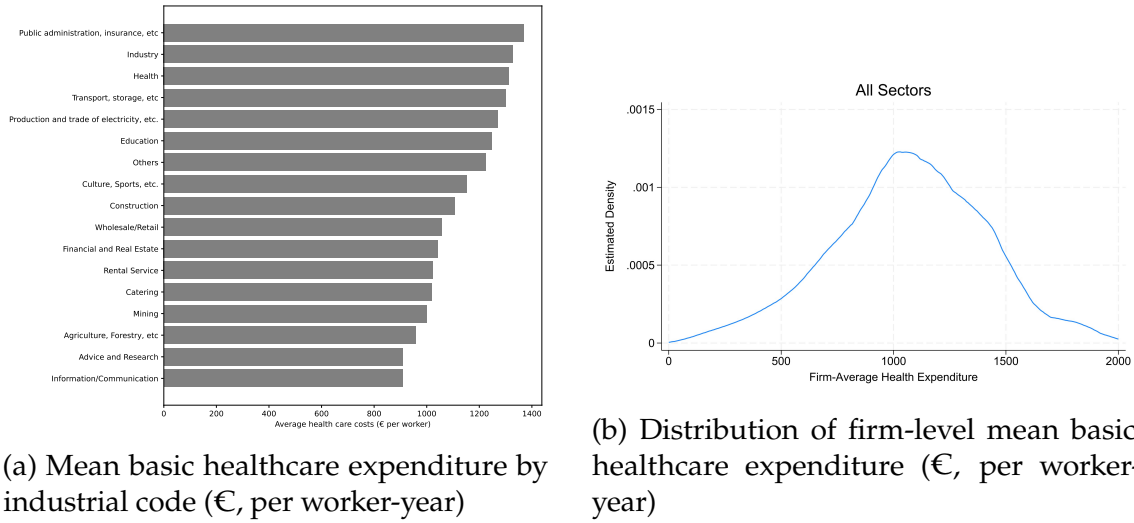


Figure 1: Raw sectoral and firm-level variation in mean annual basic healthcare expenditure

Note: Sample of male employees aged 25–65 in the connected set of firms used for the AKM estimation, observed between 2009 and 2016. Workers are restricted to full-time jobs with FTE ≥ 0.9 and at least 11 months in a given year; if an individual is employed by multiple firms within a year, the firm with the largest number of months is used. Healthcare expenditure refer to annual spending under the statutory basic health insurance package (insurer payments plus deductible) and exclude care covered only by supplementary insurance or in parallel systems (such as the defence sector, which is excluded). Sectors in panel (a) are defined from two-digit industrial codes grouped into 17 broad sectors. Panel (b) shows the cross-sectional distribution of firms' mean annual basic healthcare expenditure per worker-year, where each firm's mean is computed over all full-time male worker-year observations in the connected sample.

III.I Mover evidence on timing and pass-through

A natural next step is to examine healthcare expenditure among workers who move between firms with different average spending. The event study provides two pieces of evidence. Pre-move outcomes show whether workers moving toward firms with higher average coworker expenditure were already on different trajectories; post-move outcomes show whether and when their own spending adjusts after arrival.

Mover designs measure pass-through along observed moves: the coefficient

relates a worker’s change in spending to the origin–destination gap in coworker spending. This transition-based object is shaped by the routes movers take through the firm distribution. Even under conditions that give it a cross-sectional interpretation, that interpretation concerns variation across firm averages, not firms’ contribution to individual-level healthcare-expenditure variation (Moghani, 2026). Appendix A3 develops this distinction.

The event study and the main firm-effect estimates consequently have distinct roles. I use the event study to assess pre-trends and trace the timing and pass-through of healthcare expenditure following a change of employer. I estimate firm-effect dispersion and its contribution to individual-level expenditure variation directly with the bias-corrected two-way fixed-effects decomposition in Section V.

Here, I focus on individuals who switch employer exactly once during 2009–2016 and work with age- and year-residualised log healthcare expenditure. Let $\hat{h}_{it}$ denote the residual from a regression of log-healthcare expenditure ($h_{it} = \log(c_{it}+1)$) on age and calendar-year dummies. For each mover i , define

$$D_i \equiv \bar{\hat{h}}_{d(i)}^{(-i)} - \bar{\hat{h}}_{o(i)}^{(-i)},$$

where $\bar{\hat{h}}_{o(i)}^{(-i)}$ and $\bar{\hat{h}}_{d(i)}^{(-i)}$ are the leave-one-out mean residualised expenditure at the origin and destination firms. Thus, D_i measures how much higher or lower average residualised expenditure is at the destination firm relative to the origin.

Figure 2a relates the change in a mover’s own residualised expenditure between the pre- and post-move years to D_i nonparametrically.¹⁸ Two features stand out. First, the relationship is close to linear: workers who move to firms with higher average residualised coworker expenditure experience higher own expenditure after the move. Second, the fitted relationship has a positive intercept at $D_i = 0$: even movers between firms with similar average expenditure exhibit a modest increase in healthcare expenditure around the job change. This pattern is consistent with a common cost of moving—for example, due to job loss, or displacement—that shifts expenditure for all movers, while the slope with respect to D_i measures pass-through associated with the origin–destination gap in average coworker

¹⁸Appendix Figure A6 reports a binscatter with the linear approximation and the distribution of D_i across movers. The figure focuses on the central range of D_i , where almost all moves lie; estimates are naturally less precise in the tails.

expenditure. In turn, the approximate linearity of the relationship is consistent with the log-additive specification motivated above.¹⁹

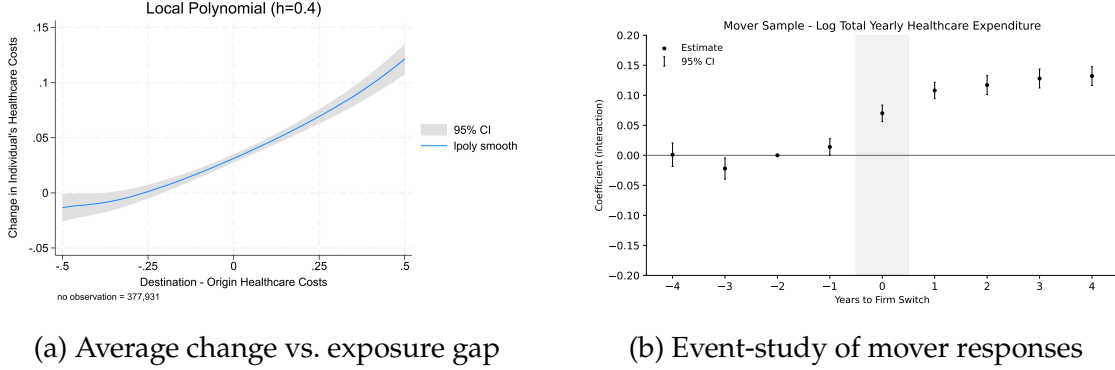


Figure 2: Workers' healthcare expenditure around moves between firms

Note: Sample of full-time male employees aged 25–65 who switch employer exactly once between 2009 and 2016 and belong to the connected set of firms. Healthcare outcomes are log total basic-package expenditures, residualised on age and calendar-year dummies. In panel (a), each point plots the average change in individual annual residualised expenditure between the years before and the years after the move against the “exposure gap” D_i , defined as the difference in leave- i -out mean residualised expenditure between destination and origin firms; the line shows the (non-parametric) fitted relationship. In panel (b), the markers show coefficients from the mover event-study specification in equation (1), where the outcome is residualised log expenditure and the regressors are event-time indicators interacted with D_i , with event time measured relative to $t = -2$ as the reference year.

To study timing more systematically, I next estimate an event-study specification that follows one-time movers from four years before to four years after the switch. Using event times $\mathcal{T} = \{-5, \dots, 5\}$ and taking $t = -2$ as the reference year, I estimate

$$\hat{h}_{it} = \eta_i + \sum_{\tau \in \mathcal{T}, \tau \neq -2} \lambda_{\tau} \mathbf{1}\{t = \tau\} + \sum_{\tau \in \mathcal{T}, \tau \neq -2} \theta_{\tau} \mathbf{1}\{t = \tau\} D_i + u_{it}, \quad (1)$$

where η_i are individual fixed effects and $\lambda_{-2} = \theta_{-2} = 0$, so all coefficients are relative to event time $t = -2$.

¹⁹By contrast, the origin–destination cell approach of Card et al. (2018) effectively imposes symmetry: the estimated effect of moving from firm a to firm b is constrained to be the mirror image of the effect of moving from b to a . In the setting of healthcare expenditure—where job loss and search themselves have health consequences—working at the individual level instead allows these common moving costs to show up as an intercept shift, while the slope with respect to D_i measures pass-through associated with the origin–destination gap in average coworker expenditure and provides a direct functional-form check. The four group split results are shown in Figure A8.

Figure 2b plots the coefficients θ_τ , which trace how pass-through evolves around the move. The event study delivers three main messages. First, the interaction coefficients, θ_τ , are small and close to zero in the years before the move, in line with no systematic pre-trends correlated with the origin–destination gap in average coworker expenditure. This argues against selection into destination firms based on differential pre-move health shocks. Second, θ_τ jumps sharply at the time of the move, showing that workers’ expenditure responds promptly. Third, only modest further adjustment follows, and the response stabilizes within the first few years. This limited adjustment shows that pass-through in the mover design occurs primarily at the move and stabilizes thereafter; it is also consistent with a time-invariant firm component in annual expenditure. Along with the approximately linear relationship in panel (a), these patterns support the log-additive two-way fixed-effects specification used in the next section. Figure 4 provides additional evidence on interpretation: moves toward firms with higher average coworker expenditure are followed by increases in medication categories associated with pain, inflammation, or related physical strain.

IV Method

To quantify how much firms contribute to variation in healthcare expenditure, I estimate a two-way fixed-effects model of log annual healthcare expenditure in the spirit of Abowd et al. (1999):

$$h_{it} = \alpha_i + \psi_{J(i,t)} + x'_{it}\beta + r_{it}, \quad (2)$$

where h_{it} is log basic-package healthcare expenditure (total paid by the insurer plus deductible paid by the individual) of worker i in year t , $J(i,t)$ indexes the firm employing i in t , and x_{it} collects time-varying observables (a 10-year split quadratic age profile by education group and calendar-year dummies). The term α_i is a worker fixed effect capturing time-invariant determinants of expenditure such as genetic endowments, early-life health, persistent preferences for care, and long-run earnings capacity. The term $\psi_{J(i,t)}$ is a firm fixed effect capturing each firm’s persistent contribution to workers’ healthcare expenditure after removing time-invariant worker heterogeneity and life-cycle and calendar-time factors, and

is interpreted below as a net workplace-related expenditure effect.

Annual basic-package healthcare expenditure is the main outcome because it is measured for every worker in every year and varies continuously rather than as a rare event. These features make it preferable, for the AKM analysis, to “hard” health outcomes such as mortality or disability, which are low-frequency tail events, observed at most once over the sample window and recorded as binary indicators. The outcome also maps naturally into a Grossman-style interpretation in which firms may affect expenditure through the workplace-specific health-depreciation rate as well as through other determinants of care use. Appendix A4 motivates the log-linear specification in a simple Grossman-style health-capital framework. Under a multiplicative structure of health investments and care-use determinants, a log specification yields additive worker and firm components, so that a given difference in firm effects implies the same percentage difference in expenditure for workers with different baseline expenditure levels, while allowing larger level effects (in euros) for higher-spending workers.

IV.I Identification and assumptions

Equation (2) relies on several assumptions.

Individual health trajectories. The residual r_{it} captures, among other things, individual-specific health trajectories over the life cycle that are not fully absorbed by the flexible age and year controls in x_{it} . Following Card et al. (2013), I assume that any remaining worker-specific trend by age has zero mean and is not systematically related to the firm effect of the worker’s employer, conditional on the included age-by-education profiles and calendar-year dummies. This rules out unobserved individual characteristics that generate differential age patterns which are themselves correlated with the firm effect at the worker’s destination. The rich age controls and the absence of pre-trends in the mover event studies (Figure 2) support this assumption: there is no evidence that workers who move toward firms with higher versus lower average coworker expenditure already display diverging healthcare trajectories in the years before the move.

Additive separability in logs. The model assumes that worker and firm effects enter additively in logs, i.e. there are no systematic worker–firm interaction terms in r_{it} . To gauge the importance of such interactions, I estimate a saturated model with contract fixed effects, replacing $\alpha_i + \psi_{J(i,t)}$ by a full set of worker×firm dummies, and compare the fit to the two-way fixed-effects model. As shown in Table 2, the R^2 increases from 0.599 to 0.657 when moving from the AKM to the saturated specification, but the adjusted R^2 rises only modestly, from 0.516 to 0.532, despite an increase of 56% in the number of parameters.

Figure 3 plots the mean residuals $\hat{r}_{it}$ from the AKM model by deciles of the estimated worker and firm effects. While the variance of $\hat{r}_{it}$ in the sample is close to one, the averages are close to zero across almost all cells, with small deviations only in the cell combining the lowest worker and firm deciles, where truncation at zero and extensive-margin responses are most relevant. This pattern suggests that firm effects may differ somewhat for individuals at the very bottom of the expenditure distribution, but that additive separability in logs is a good approximation for the large majority of workers. Additionally, in Section V, I re-estimate firm effects separately for subsamples defined by education (blue- vs. white-collar), income, and baseline expenditure and show that the dispersion of firm effects and their correlation across subsamples are high, supporting the interpretation of a common firm component.

A further functional-form check comes from the mover design in Figure 2a. The non-parametric relationship between the change in an individual’s residualised log expenditure and the exposure gap D_i is approximately linear, with a common positive intercept at $D_i = 0$. The approximate linearity with a constant intercept is therefore consistent with the log-linear, additively separable structure in Equation (2).

Time-invariant worker and firm effects. I treat α_i and ψ_j as constant over the 2009–2016 period. Any transitory firm shocks or worker-specific changes in long-run health are absorbed into r_{it} . Given the relatively short panel and the limited dynamic adjustment after job moves shown in Figure 2b, this appears to be a reasonable approximation.²⁰

²⁰Allowing fully time-varying firm-specific health effects would require a longer panel and richer mobility than the data permit.

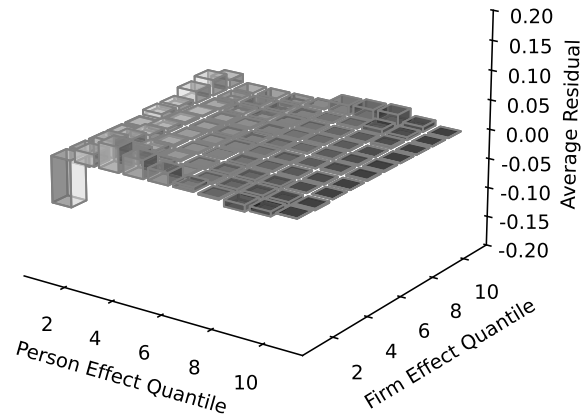


Figure 3: Mean residuals of the two-way fixed-effects model (Eq. 2) by decile of firm and individual fixed effects.

Note: Sample of full-time male employees aged 25–65 in the connected set of firms, observed between 2009 and 2016. The figure plots mean residuals from the two-way fixed-effects regression in Eq. 2, where the dependent variable is log annual basic-package healthcare expenditure, $h_{it} = \log(c_{it} + 1)$, and the controls are worker and firm fixed effects, age-by-education profiles, and calendar-year dummies. Residuals are averaged within 10×10 cells defined by deciles of the estimated worker and firm fixed effects (decile 1 = lowest, decile 10 = highest). The variance of the residuals is approximately 1, so the cell means shown in the figure represent small deviations relative to the overall idiosyncratic dispersion.

Model	Two-Way Fixed Effects		Saturated with Job ID	
	R^2	Adjusted R^2	R^2	Adjusted R^2
	0.599	0.516	0.657	0.532
No. of Parameters	2,901,488		4,527,516	
# Person–Year Observations: 16,976,071				

Note: Sample of full-time male employees aged 25–65 in the connected set of firms, observed between 2009 and 2016 (# person–year observations = 16,976,071). The dependent variable is log annual basic-package healthcare expenditure (insurer payments plus deductible), $h_{it} = \log(c_{it} + 1)$. The two-way fixed-effects model (Eq. 2) includes worker and firm fixed effects and controls for age–by–education profiles and calendar-year dummies. The saturated model replaces worker and firm fixed effects with job-ID (worker×firm) fixed effects while keeping the same controls. The table reports R^2 , adjusted R^2 , and the total number of estimated parameters in each specification.

Table 2: Two-Way Fixed Effects vs. Saturated (Job-ID) Model

Selective mobility and pre-trends. A central concern is that workers may move to firms with different health environments in anticipation of future health shocks, violating the implicit exogeneity condition that, conditional on α_i , and age–education patterns and time dummies, the firm effect is unrelated to unobserved changes in health. I provide two pieces of evidence against strong anticipatory mobility. First, the mover event study in Figure 2b, which interacts event-time indicators with the exposure gap D_i defined in Section III, shows that the coefficients θ_τ are flat and close to zero in the years before the move, while expenditure differentials open up sharply at the time of the move and then remain roughly constant. Second, Table A7 shows that excluding workers above age 45, for whom future health declines are more common and predictable, leaves the estimated dispersion of firm effects robust. This age split is consistent with life-cycle health-capital models in which individuals are close to full health throughout most of their prime working years and more rapid health deterioration sets in mainly at older ages (Galama and Van Kippersluis, 2019). Moreover, existing evidence for the Netherlands documents substantial worker and job dynamics, with job-to-job flows an important component of labour reallocation (e.g. Adema et al., 2025), making it plausible that many employer switches reflect career progress or job matching rather than responses to anticipated health problems. Taken together, these patterns suggest that selective mobility based on unobserved, destination-specific health trends is limited.

Idiosyncratic shocks. The remaining component of r_{it} reflects idiosyncratic, time-varying shocks to healthcare use that are not captured by the fixed effects or observed covariates. I assume that these shocks have zero mean and are not systematically related to firm assignment or the decision to move, conditional on α_i , ψ_j , and x_{it} . The large variance of r_{it} and the small average residuals across worker–firm cells in Figure 3 support this view that most of the residual variation reflects noise rather than systematic departures from the model structure.

IV.II *Object of interest and inequality interpretation*

The main object of interest is the dispersion of firm effects,

$$\sigma_\psi^2 \equiv \text{Var}(\psi_j), \quad \sigma_\psi = \text{sd}(\psi_j).$$

A move from firm A to firm B changes a worker’s expected log expenditure by $\Delta\psi = \psi_B - \psi_A$, holding α_i and x_{it} fixed, so that

$$\frac{\mathbb{E}[c_{it}^{(B)}]}{\mathbb{E}[c_{it}^{(A)}]} = \exp(\Delta\psi).$$

In particular, expected expenditure for the same worker differs by a factor $\exp(\sigma_\psi)$ between firms whose effects are one standard deviation apart. I therefore report σ_ψ and $\exp(\sigma_\psi) - 1$ as the primary measures of how much firms can shift individual healthcare expenditure.

I use this absolute measure because it answers the substantive question directly: how much expected healthcare expenditure differs across firm environments for the same worker. A variance share instead scales firm effects by total individual-level variation, much of which reflects short-lived and idiosyncratic healthcare shocks, and can therefore obscure economically meaningful workplace differences (Kline, 2024). The mover analysis answers a complementary question—how much of the observed origin–destination gap passes through along the transitions seen in the data—rather than directly estimating the dispersion of firm effects across the workforce or their contribution to individual-level expenditure variation (Moghani, 2026). I therefore estimate σ_ψ directly and translate it into changes in workers’ expected expenditure.

This choice aligns closely with how health disparities are typically measured in the literature. For wages or incomes, univariate inequality indices (such as the variance or the Gini coefficient) have a natural welfare interpretation because the variable is monetary, cardinal, and directly comparable across individuals. For health expenditure, by contrast, such univariate indices are used much less frequently, in part because a large share of the dispersion reflects idiosyncratic biological risk and random health shocks²¹, and there is no clear normative benchmark for what constitutes “acceptable” or unavoidable dispersion.²² Instead, empirical work typically studies socioeconomic-related health disparities using bivariate measures based on the joint distribution of health and another variable such as income, education, or socio-economic status—for instance concentration indices, slope indices of disparities, or health gaps across income, education, or regional groups (e.g. [Kakwani et al., 1997](#); [Wagstaff, 2002](#); [Wagstaff and Doorslaer, 2004](#); [O’Donnell et al., 2015](#); [Finkelstein et al., 2016](#)). These measures answer questions of the form “by how much is health worse for individuals in disadvantaged positions?” rather than “what fraction of the total variance is explained by a given factor?”. In this sense, σ_ψ provides a firm-level analogue to these bivariate gradient measures, with the firm effect replacing income or education and healthcare expenditure replacing the health outcome.

IV.III *Limited mobility and leave-out variance correction*

Equation (2) can be estimated by OLS. However, variance components computed from the plug-in fixed-effect estimates, $\{\hat{\alpha}_i, \hat{\psi}_j\}$, are biased when worker mobility is limited. In employer networks where only few workers move between subsets of firms, each fixed effect is estimated with substantial noise even if it is unbiased individually. When these noisy estimates enter quadratically, as in $\widehat{\sigma_\psi^2}$, the variance of the estimation error inflates the quadratic form and leads to systematic bias

²¹In classic earnings data, regressions of log earnings on basic human-capital and labor-supply variables (schooling, experience) explain about 30% of the cross-sectional variance ($R^2 \approx 0.3$; see [Mincer \(1974\)](#)), whereas analogous models for annual healthcare expenditure using only age, gender, region, and source of income explain on the order of 6% of the variance ($R^2 \approx 0.06$; see [Van Kleef et al. \(2013\)](#)).

²²See, for example, [Wagstaff et al. \(1991\)](#), [Kakwani et al. \(1997\)](#), [Wagstaff and Doorslaer \(2004\)](#), [Erreygers and Van Ourti \(2011\)](#), and [Schlottheuber and Hosseinpoor \(2022\)](#), who distinguish overall from socioeconomic-related health disparities and show that empirical work and monitoring practice typically focus on the latter using concentration-type and related bivariate indices.

(Andrews et al., 2008; Jochmans and Weidner, 2019; Kline et al., 2020; Bonhomme et al., 2023). Intuitively, the plug-in variance of firm effects is biased upward because it adds true dispersion and noise dispersion. This problem is often referred to as *limited mobility bias*.²³

To address this, I use the leave-out variance estimator proposed by Kline et al. (2020) (KSS). Their approach provides unbiased estimators for variance and covariance components of interest in linear models without imposing homoskedasticity or additional structural restrictions. Formally, any variance or covariance term can be written as a quadratic form $\theta = \nu' A \nu$, where ν stacks all model parameters and A is a known symmetric matrix. The naive plug-in estimator $\hat{\theta}_{PI} = \hat{\nu}' A \hat{\nu}$ has a bias that can be expressed as a weighted sum of observation-specific error variances. KSS shows how to estimate these error variances in a leave-one-out fashion and subtract the implied bias, yielding a bias-corrected estimator $\hat{\theta}_{KSS}$.

In practice, I implement the KSS correction at the worker–firm match level. Following Kline et al. (2020), I first partial out the observed covariates x_{it} by estimating Equation (2) and working with the residualised outcome $h_{it} - x'_{it}\hat{\beta}$. I then compute bias-corrected estimates of σ_{ψ}^2 using the leave-one-out formulas of Kline et al. (2020), treating an entire worker–firm match as the leave-out unit to accommodate serial correlation within matches. Because recomputing the fixed effects separately for each leave-out observation is infeasible in a large panel, I follow KSS and use their analytic leave-out formulas and random-projection approximation based on Achlioptas (2003) to obtain computationally tractable estimates.²⁴ Appendix A5 provides the quadratic-form representation of the variance components and details of the implementation; see Kline et al. (2020) for a full derivation of the leave-out estimator. As a robustness check, I also re-estimate the variance components on a more strongly connected subset of firms; as expected, the plug-in bias is smaller but not negligible, and the KSS correction still reduces the estimated dispersion of firm effects (Table A15). These strongly connected firms are larger and more selected than the average employer and are therefore not representative of the full economy.

²³The same logic applies to other quadratic terms such as $Cov(\alpha_i, \psi_j)$; these are corrected using the same leave-out procedure.

²⁴All AKM and leave-out computations are implemented in R using the package `LeaveOutKSS` developed by the author, which adapts the original Matlab routines of Kline et al. (2020) available at [GitHub: LeaveOutKSS-R](#).

V Results

Firms differ substantially in the healthcare expenditure they induce among otherwise similar workers. Table 3 reports KSS-corrected variance and covariance components for log annual healthcare expenditure h_{it} in Equation (2). The variance of firm effects is $\text{var}(\psi_j) = 0.027$, implying $\sigma_\psi \approx 0.164$ log points. A one-standard-deviation difference in firm effects corresponds to a difference of $\exp(\sigma_\psi) - 1 \approx 17.8\%$ in expected annual basic healthcare expenditure for the same worker. The worker-firm correlation is close to zero ($\rho(\alpha_i, \psi_j) = -0.018$), showing limited sorting on time-invariant expenditure.

To benchmark magnitudes, I compare the workplace gradient to the well-known gradients by wealth, income, and education that are standard in the health-disparities literature.²⁵ Within the sample, the change in (age- and year-adjusted) log healthcare expenditure from the bottom to the top quintile of the wealth distribution is about 0.26; for income, the corresponding difference is 0.29; and comparing workers with elementary or lower education to those with a master’s or doctorate yields a difference of 0.40 (Table A2). These gaps correspond to approximately 29%, 33%, and 49% higher annual basic healthcare expenditure, respectively. Using the KSS-corrected dispersion of firm effects and a normal benchmark, the implied difference in expected expenditure between otherwise identical workers employed at firms in the bottom and top quintiles of the firm-effect distribution is about 0.46 log points, i.e. roughly 59% higher annual basic healthcare expenditure.²⁶ In this sample, the implied workplace gradient in expenditure is therefore comparable in magnitude to the observed gradients by wealth, income, and education.²⁷

²⁵Highest completed education is used to define education; household wealth and total personal income are measured in 2009, before the main analysis period, to reduce concerns about reverse causality. These SES gradients are descriptive correlations rather than causal effects, and I report them only as a familiar benchmark for the magnitude of the firm-induced expenditure gradient.

²⁶The “top–bottom quintile gap” is the difference between the mean firm effects in the top and bottom quintiles of the worker–year weighted distribution. I translate the leave-one-out bias-corrected variance estimate of Kline, Saggio, and Sølvsen (KSS; Kline et al., 2020), $\widehat{\sigma}_{\psi,\ell}^2$, into this gap using a normal approximation to the firm-effect distribution, which implies about 0.46 log points.

²⁷The sample is positively selected in terms of socioeconomic characteristics: it consists of full-time male workers with relatively high income and wealth by construction (Figure A9), which likely underestimates the SES gradients in the Netherlands. In the same institutional context, Loef

For completeness, Table 3 also reports variance shares: firm effects account for only about 1% of the total variance in h_{it} at the individual-year level. As discussed in Section IV, this variance share is not the most informative summary statistic, because the denominator is dominated by large idiosyncratic individual-level shocks—such as cancers, accidents, and chronic conditions—that no employer can plausibly control and that are at least partly inherently random.

Statistic	KSS Estimates
Log Health Exp. Var.: $\text{var}(h_i)$	2.617 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.027 (1.03 %)
Employee FE Var.: $\text{var}(\alpha_i)$	1.516 (57.92 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	−0.004
Correlation $\rho(\alpha_i, \psi_j)$	−0.018
Explained share: workers & firms	0.587
No. Movers	1,225,799
No. Firms	132,820
No. of Observations	16,976,071
Average Health Exp. : $\text{mean}(h_i)$	5.608

Notes: Components are estimated via the leave-one-match-out Kline–Saggio–Sølvsten (KSS) correction (Kline et al., 2020) using random projections (50 simulations). Outcome is log healthcare expenditure per calendar year; sample spans full-time employees, 2009–2016. Shares in parentheses are relative to $\text{var}(h_i)$.

Table 3: Variance Decomposition of Log Healthcare Expenditure

V.I Firm Fixed Effects and Health

Several results suggest that the estimated firm effects are more consistent with workplace-linked differences in health risk than with insurance design, income-related care use, or short-run utilisation incentives. Adding annual gross contract earnings leaves both the dispersion and the ranking of firm effects essentially unchanged (Table A4), suggesting that the estimates do not simply proxy for pay-related differences in healthcare use. Dispersion remains sizeable among workers with high baseline expenditure, who typically exceed the deductible

et al. (2021) show that healthcare expenditure is between 50% and 150% higher among individuals in the lowest education or income group than among those in the highest group across age bands. This suggests that, in the full population, SES-related gradients are larger than the 30–39% gaps observed in my sample.

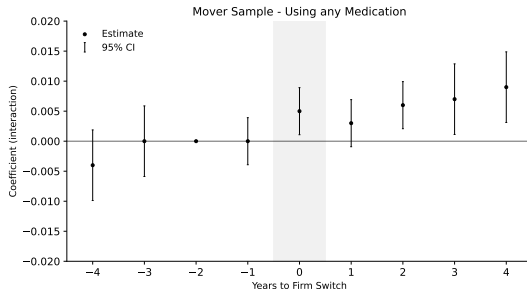
and face a zero marginal price for additional basic care, and firm rankings are aligned across high- and low-baseline expenditure groups (cross-sample slope 0.60) (Table A5). The same alignment holds when splitting by hourly wages (Table A6). Category-specific estimates point in the same direction. GP spending, which is fully covered and deductible-exempt, exhibits a firm-effect standard deviation close to that for total expenditure (Table A9). These patterns are difficult to reconcile with employer-specific insurance design, prices, or income-related care use as the main drivers of the firm effects.

To provide complementary evidence on whether the expenditure changes observed around moves reflect health-risk channels rather than broad differences in utilisation or diagnosis intensity, I examine changes in medication use around firm moves using Equation (1). I consider both an indicator for any medication use and indicators for three subgroups: pain, anti-inflammatory, and muscle relaxant medications; cardiometabolic medications; and cancer-related medications. Consistent with the rise in healthcare expenditure among workers moving toward firms with higher average coworker expenditure (Figure 2), Figure 4 shows an increase in overall medication use following such moves. This increase is not accompanied by changes in cardiometabolic or cancer-related medications, suggesting that differential screening or referral for previously undiagnosed conditions is unlikely to explain the expenditure pass-through documented in the mover design. Instead, the increase is concentrated in pain, anti-inflammatory, and muscle relaxant medications, a pattern more consistent with workplace strain, injury, or work-related stress than with broader changes in diagnosis intensity.

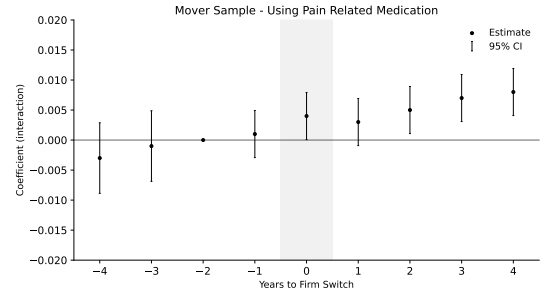
To further study the link between the estimated firm effects to health outcomes, I study two long-run measures observed in 2019. The first is an indicator for death by 2019, which captures cumulative severe health deterioration. The second is an indicator for disability receipt in 2019,²⁸ which captures persistent work-limiting health conditions. I relate these outcomes to firm effects on healthcare expenditure, conditioning on baseline demographics, pre-period expenditure, and origin-firm characteristics.

To prevent the same workers' healthcare histories from determining both the

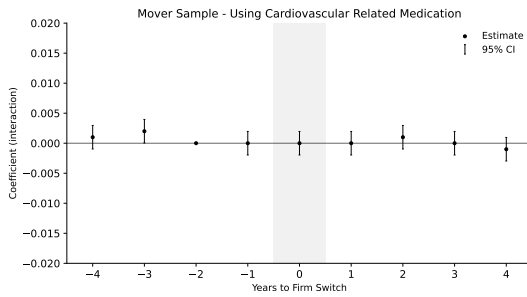
²⁸Disability status is a truncated outcome: by construction, only individuals who are alive and not on other competing benefits or social welfare programmes can be on disability.



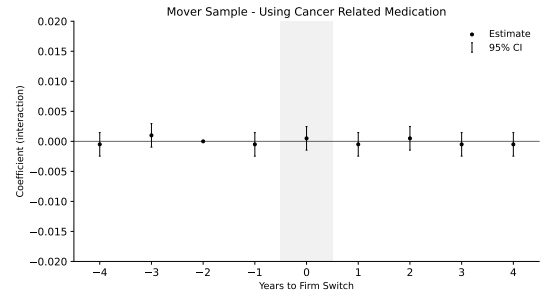
(a) Any medication



(b) Pain, anti-inflammatory, and muscle relaxant medications (ATC: N02, M01, M03)



(c) Cardiometabolic (ATC: C03, C07, C08, C09, C10, A10, B01)



(d) Cancer-related (ATC: L01, L02, L03)

Figure 4: Medication use around moves between firms

Note: Sample of full-time male employees aged 25–65 who switch employer exactly once between 2009 and 2016 and belong to the connected set of firms. Each panel reports coefficients from the mover event-study specification in equation (1), where the outcome is an indicator for using (having any reimbursed dispensing of) the medication group in a calendar year, residualised on age and calendar-year dummies. Regressors are event-time indicators interacted with the exposure gap D_i , defined as the difference in leave- i -out mean residualised total healthcare expenditure between destination and origin firms; event time is measured relative to $t = -2$ (reference year). Markers show point estimates and vertical bars show 95% confidence intervals. Medication groups are defined by ATC prefixes: pain, anti-inflammatory, and muscle relaxant medications (N02 analgesics, M01 anti-inflammatory/antirheumatic products, M03 muscle relaxants; consistent with musculoskeletal strain and acute injury-related treatment), cardiometabolic medications (C03 diuretics, C07 beta blockers, C08 calcium channel blockers, C09 RAAS agents, C10 lipid modifiers, A10 diabetes drugs, B01 antithrombotics), and cancer-related medications (L01 antineoplastic agents, L02 endocrine therapy, L03 immunostimulants).

firm-effect measure and their downstream outcomes, I follow [Kline \(2025\)](#) and randomly split workers into two subsamples. I estimate the two-way fixed-effects model separately in each,²⁹ keep firms observed in both, and assign each worker the

²⁹See Table A17 for the KSS variance estimate within each subsample.

	Mortality in 2019		Disability in 2019	
	(1)	(2)	(3)	(4)
Average sample-split firm-effect exposure	0.00111*** (0.00020)	0.00212*** (0.00020)	0.00377*** (0.00036)	0.00779*** (0.00035)
Worker FE control	No	Yes	No	Yes
Migration background controls	Yes	Yes	Yes	Yes
YOB × education FEs	Yes	Yes	Yes	Yes
Postal code FEs	Yes	Yes	Yes	Yes
<i>N</i>	2,485,666	2,485,666	2,432,508	2,432,508
<i>R</i> ²	0.0417	0.0483	0.0445	0.0754

Notes: Each column reports a linear probability model estimated. The regressor is the worker-level panel average of the sample-split firm effect on log healthcare expenditure, computed over the years the worker is observed between 2009–2016. All specifications include migration background indicators, postal code fixed effects, and year-of-birth × education-category fixed effects. Columns (2) and (4) additionally control for the worker fixed effect estimate. Robust standard errors in parentheses. *** $p < 0.01$.

Table 4: Average Estimated Firm-Effect Exposure and Subsequent Mortality and Disability (2019)

demeaned firm effect estimated in the other subsample. Table 4 shows that workers with greater exposure to firms with higher estimated healthcare-expenditure effects are more likely to be on disability and more likely to have died by 2019, conditional on migration background, postal-code fixed effects, and year-of-birth × education fixed effects. The association becomes larger after controlling for the worker fixed-effect estimate, reinforcing that it is not explained by time-invariant differences in baseline health.

I next use firm closures as an additional source of exogenous mobility. When a firm exits, all workers separate in the same year and move to destination firms with different estimated health-cost effects. I focus on workers observed from the start of the panel and followed continuously, allowing me to trace healthcare trajectories around displacement. Following the displacement literature (e.g., [Halla et al. \(2020\)](#)), I define closure-induced job loss as a transition in which the employer identifier disappears from the panel and take the closure year as the last year the firm is observed. Figure A11 shows that, after reemployment, workers' expenditure shifts in the direction of the destination firm's average coworker expenditure.

Table 5 reports regressions of disability and mortality in 2019 on the sample-

split estimate of the destination firm’s health-cost effect, controlling for baseline demographics, pre-closure average healthcare expenditure, and the corresponding sample-split estimate of the origin firm’s health-cost effect.^{30,31} The magnitudes are economically meaningful. Using the baseline AKM dispersion ($\sigma_\psi = 0.164$), reemployment at a destination firm one standard deviation higher in the health-cost firm-effect distribution is associated with a 0.22 percentage point higher probability of disability in 2019 and a 0.08 percentage point higher probability of death, roughly 6% of the respective sample means.³² Figure 5 provides a complementary visualization: within broad origin bins, (residualized) mortality and disability are higher for workers reemployed at destination firms with high estimated health-cost effects than for those reemployed at destination firms with low estimated health-cost effects.^{33,34}

To characterise what the firm effects capture within healthcare expenditure and related health outcomes, I decompose expenditure by care category and relate workers’ exposure to estimated firm effects on healthcare expenditure to cause-specific mortality. Dispersion in firm effects is present across care domains, and it is largest for hospital and specialist somatic care, smaller for GP services and medication—suggesting that firm effects are present for both severe and more routine care categories—and modest for specialised mental healthcare (Table A9). In the mortality data, exposure to firms with higher estimated healthcare-expenditure effects is most strongly associated with subsequent cancer

³⁰See Table A10 for summary statistics of the closure sample.

³¹To reduce noise in the destination measure, I restrict the sample to destination firms above the 25th percentile of the firm-size distribution (at least 75 FTE workers). Origin and destination firm effects are highly correlated ($\rho \approx 0.64$), so specifications with origin-firm fixed effects rely on limited within-origin dispersion and are correspondingly less precise and less robust. Results with origin fixed effects are similar but less precisely estimated.

³²These effect sizes use the preferred estimates in column (2) of Table 5 and the dependent-variable means reported in the table.

³³Figure A12 shows a similar relationship using a finer 5×5 (quintile-by-quintile) grouping.

³⁴As a placebo, I re-estimate the preferred specification using average pre-move healthcare expenditure as the outcome and the same covariate set as in Table 5. The destination-firm coefficient is not positive; if anything, it turns negative once I condition on the origin firm’s health-cost effect. Because origin and destination firm effects are highly correlated in the closure sample, placebo estimates that condition on origin-firm fixed effects should be interpreted cautiously. In any case, the lack of a positive relationship between destination fixed-effect and pre-move healthcare expenditures reduces concerns that the documented relationship reflects pre-existing differences in health (Table A11).

	(1)	(2)	(3)
<i>Outcome measured in 2019</i>			
Panel A: Disability in 2019			
Destination firm effect (sample-split)	0.0078*** (0.00133)	0.0137*** (0.00192)	0.0086*** (0.00292)
Mean of dep. var.		0.03929	
<i>N</i>	206,305	206,305	204,012
<i>R</i> ²	0.066	0.066	0.116
Panel B: Mortality in 2019			
Destination firm effect (sample-split)	0.0033*** (0.00078)	0.0047*** (0.00113)	0.0034* (0.00173)
Mean of dep. var.		0.01313	
<i>N</i>	209,235	209,235	206,945
<i>R</i> ²	0.048	0.048	0.083
Baseline controls & pre-move postal-code FEs	Yes	Yes	Yes
Origin firm-effect level (linear control)	No	Yes	No
Origin-firm fixed effects	No	No	Yes

Notes: Sample of displaced male workers following firm closures, restricting destination firms to those above the 25th percentile of firm size (at least 75 employees). The key regressor is the sample-split destination-firm effect on log healthcare expenditure. Column (2) additionally controls for the sample-split origin firm-effect level; column (3) includes origin-firm fixed effects. Baseline controls include pre-move average healthcare expenditure and demographics (as in the text) and pre-move postal-code fixed effects. Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: Firm Closures: Destination Firm Health-Cost Effects and Long-Run Outcomes (Continuous Measure)

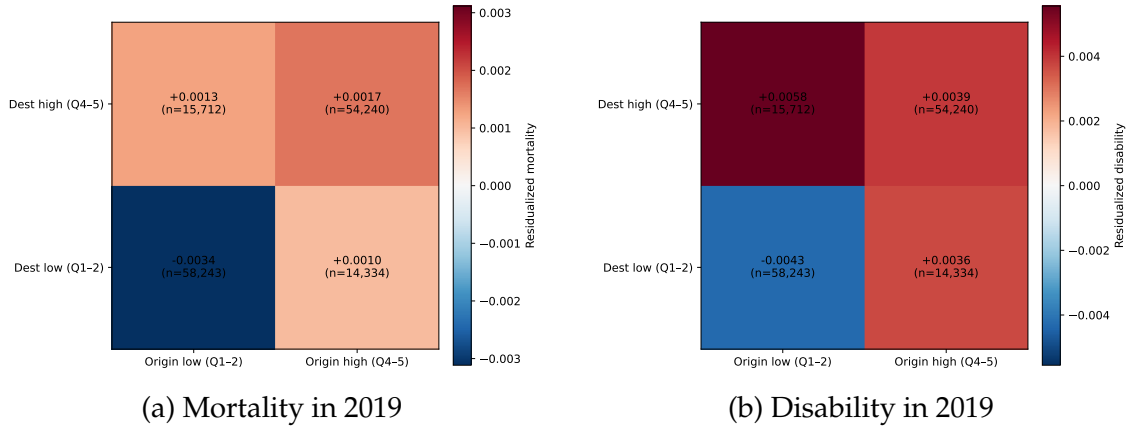


Figure 5: Long-run outcomes by origin and destination firm-effect bins after firm closures

Note: Sample of displaced male workers following firm closures, restricting destination firms to those above the 25th percentile of firm size (at least 75 employees). Each panel plots binned means of residualized outcomes, where residuals are obtained from the same control set used in Table 5 (baseline demographics, pre-move average healthcare expenditure, and pre-move postal-code fixed effects). The horizontal and vertical axes group workers by the sample-split origin and destination firm health-cost effects, respectively, into coarse bins (low vs. high). Cell annotations report the number of observations used to compute each binned mean. Appendix Figure A12 shows the same relationships using a more granular 5×5 (quintile-by-quintile) binning.

deaths, whereas links with injury-related³⁵, cardiovascular, and other causes are smaller and estimated less precisely (Table A12)³⁶. These patterns, along with the increase in the use of pain, anti-inflammatory, and muscle relaxant medications (Figure 4), suggest that firms' contributions to healthcare expenditure are multidimensional and appear most closely linked to somatic and longer-run physical risks.

V.II *Heterogeneity Across Sectors, Firms, and Workers*

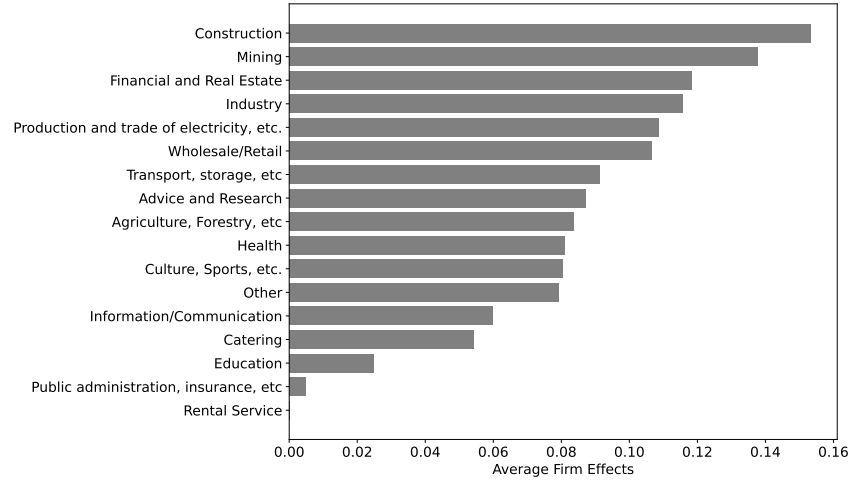
Because production technologies differ sharply across sectors, gaps between, for example, construction and finance in terms of induced health expenditure, mostly reflect what firms produce and leave limited scope for policy short of changing the activity mix. The more relevant question is how much firm-effect dispersion remains among firms that ostensibly do the same thing. Figure 6 plots observation-weighted sector means of the workplace fixed effects. As expected, traditionally high-risk sectors such as mining, construction, and manufacturing are among the worst performers, and the financial sector—where work is less physically demanding but characterised by high pressure—also stands out with large adverse effects on employee healthcare expenditure, consistent with both physical and psychosocial channels. Using 17 broad sector classifications, sector differences account for only about one sixth of the dispersion in firm health effects. When I replace these broad groups with detailed, highly granular industrial codes, industrial-code differences explain just under half of the dispersion,³⁷ so more than half of the dispersion remains within narrowly defined industrial-code cells.

³⁵Workplace mortality due to injuries is rare; in the Netherlands, annual fatal occupational accidents typically ranged between 23 and 45 cases in 2014–2020 [Centraal Bureau voor de Statistiek \(CBS\) \(2022\)](#).

³⁶Cause-specific mortality is rare in the sample, so estimates for several causes are naturally imprecise. Cancer deaths are the most common cause-specific category in these data, whereas accidental deaths and other cause-specific deaths are much less frequent.

³⁷The sector and industrial-code shares are computed in the subsample of firms that can be linked to the business registry, which removes about 12% of firms (around 1.4% of person–year observations). In this subsample, the overall variance of firm effects is smaller ($\sigma_\psi \approx 0.11$ instead of 0.16; see Table A13), so the same sector differences translate into somewhat larger variance shares than in the full connected sample. Because some SBI industrial-code cells contain relatively few firms and workers, the between-code component may also be slightly upward biased; the reported industrial-code shares should therefore be interpreted as upper bounds, and the within-code component as a conservative estimate of within-industry firm-effect dispersion.

Thus, while some activities are clearly more hazardous than others, a large share of firm-effect dispersion remains among firms that do very similar work, pointing to firm-specific practices—and hence to scope for firm-level policy interventions—as a central driver.



Notes: The figure plots observation-weighted sector means of the estimated workplace fixed effects ψ_j from Equation 2. Higher bars indicate larger average impacts on employee healthcare expenditure. Sectors are ordered by magnitude. Because each sector aggregates many firms, sampling error in sector means is expected to be small. Sectors are defined using firms' two-digit SBI industrial codes.^a As the firm effects are identified only up to a constant, I normalise the average effect in the Rental Services sector to zero; all other sector means should be interpreted as differences relative to this baseline.

^aOverzicht Standaard Bedrijfsindeling (SBI) codes are Dutch industrial codes used to classify firms by their main industry.

Figure 6: Average Sectoral Fixed Effects on Total Healthcare Expenditure

To characterise which firms drive the estimated health-cost effects, I link the AKM firm effects to observable firm characteristics from the business registry and to wage premia from an AKM model of earnings.^{38,39} Figure 7 reports these relationships as simple bivariate regressions of the firm health effect on each

³⁸This follows the compensating-differentials tradition (e.g. Rosen, 1986), and connects to recent firm-level evidence on wage premia and amenities (e.g. Sorkin, 2018; Sockin, 2022; Lamadon et al., 2022).

³⁹Firm wage premia are estimated using the same specification as in Equation (2), replacing log healthcare expenditure with log annual earnings and restricting to full-time workers; see Table A3. Hourly wages are approximated by dividing annual earnings by a full-time schedule. Standard errors use the leave-one-out procedure of Kline et al. (2020), which is robust to heteroskedasticity and to correlation induced by common estimation error in the firm effects.

characteristic (one at a time). Firms with larger workforces, higher female and migrant shares, and older workforces have lower health-cost effects, while firms with more overtime have higher health-cost effects. Firms that employ an in-house company doctor are also associated with lower health-cost effects. Finally, the ranking of estimated healthcare-expenditure effects is positively related to wage premia: firms with higher estimated healthcare-expenditure effects pay higher wage premia. These patterns are consistent with compensating differentials—wages partially offset workplace conditions that are costlier for health—although the wage premium also correlates with overtime and firm size, so the association does not by itself measure how much of the health risk is compensated.

Wage premia between and within industries. Table 6 looks at the wage-premium association more closely by asking whether it holds among firms in the same industry. Across all firms, a one log-point higher health-cost effect is associated with a 0.020 log-point higher wage premium (column 1), so a firm one standard deviation higher in the health-cost distribution ($\sigma_\psi = 0.164$) pays a wage premium roughly 0.3% higher. The association is similar among firms with a registry industry code (column 2). Within detailed industries, however, it largely disappears: adding fixed effects for the 898 SBI codes reduces the slope to 0.0007 (column 3). Read at face value, the wage premium attached to health-cost effects arises between industries—workers in industries whose firms generate higher healthcare expenditure are paid more—whereas firms that generate higher expenditure than others in the same line of business do not pay correspondingly higher premia. Since more than half of the dispersion in health-cost effects lies within detailed industries, this pattern suggests that much of the firm-level component of health risk carries no compensating wage premium.

Two caveats apply, and the slopes in Table 6 should be read as a qualitative pattern rather than as magnitudes. First, wage premia reflect rent sharing and work intensity as well as compensation for disamenities, so neither slope measures how much of the health risk is compensated. Second, both firm effects are estimated on the same workers, so their estimation errors are correlated, and the health-cost effect on the right-hand side is measured with error. Measurement error attenuates the within-industry slope more than the pooled slope, because the signal variance

of firm effects is smaller within industries while the noise variance is not.⁴⁰

	(1) All firms	(2) Firms with SBI code	(3) Firms with SBI code
Firm health-cost effect $\hat{\psi}_j$	0.0201 (0.0001)	0.0229 (0.0001)	0.0007 (0.0001)
Detailed industry (SBI) fixed effects	No	No	Yes
Observations (worker-years)	16,976,071	14,815,016	14,815,016
R^2	0.002	0.003	0.462

Notes: Worker-year regressions of the estimated firm wage premium $\hat{\psi}_j^w$ (from the wage AKM model; Table A3) on the estimated firm health-cost effect $\hat{\psi}_j$ (Equation 2), for full-time male workers in the connected set, 2009–2016. Columns 2 and 3 restrict the sample to firms with a valid SBI code; column 3 adds fixed effects for the 898 detailed SBI codes. Standard errors in parentheses are conventional worker-year standard errors and understate uncertainty: they treat the estimated firm effects as data and ignore estimation error in $\hat{\psi}_j$ and $\hat{\psi}_j^w$, which is correlated across the two outcomes because both are estimated on the same workers. The slopes are attenuated by measurement error in $\hat{\psi}_j$, more so within industries; see the text.

Table 6: Firm Wage Premia and Firm Health-Cost Effects, Between and Within Industries

Subsample analyses show that firms’ health-cost effects are largely common across workers, but with meaningful heterogeneity in magnitude. Re-estimating the AKM model separately by age, income, baseline expenditure, and collar type, the estimated firm effects line up closely across groups: firms with higher estimated healthcare-expenditure effects for one group tend also to have higher effects for others, as reflected in the cross-sample slope regressions of firm effects reported in Appendix Tables A5–A8. At the same time, the dispersion of firm effects is systematically larger for younger, lower-income, low-spending, and blue-collar workers, consistent with these groups being more exposed to channels such as physical hazards, work intensity, or tighter budget and time constraints.⁴¹

⁴⁰Under classical measurement error, the plug-in and bias-corrected variances of firm effects (0.113 and 0.027; Tables 3 and A15) imply that the pooled slope is attenuated by a factor of about four and the within-industry slope by roughly twice as much, which is far from the thirty-fold difference between columns 1 and 3. Correlated estimation error between the wage and health-cost effects, by contrast, would affect the pooled and within-industry slopes similarly. A split-sample design—health-cost effects estimated from one random half of the workforce, instrumented with those from another half, and wage premia estimated from a third—removes both problems and will replace these estimates in a subsequent version.

⁴¹Blue- and white-collar status is proxied from fields of study using a text-based classification.



Notes: The figure reports OLS coefficients from separate bivariate regressions where the dependent variable is the estimated firm fixed effect in (log) healthcare expenditures. Each coefficient estimates the univariate regression coefficient of the firm health-cost effect on the indicated firm characteristic. Points show point estimates; whiskers show 95% confidence intervals computed using KSS leave-one-out standard errors (Kline et al., 2020), which are robust to heteroskedasticity and to correlation induced by common estimation error in the firm effects. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Figure 7: Correlates of Firm Fixed Effects in Healthcare Expenditure

V.III Robustness Checks

A key identifying assumption is that workers do not change employers in anticipation of unobserved health deterioration. The mover event studies in Figures 2 and A8 show no systematic pre-trends associated with the origin–destination gap in average coworker expenditure. To further reduce scope for health-driven mobility, Table A7 re-estimates the AKM model and corresponding variance decomposition for workers aged ≤ 45 , a life-cycle phase in which health and healthcare use are typically stable (Galama and Van Kippersluis, 2019). The KSS-corrected dispersion of firm effects is close to the baseline estimate in Table 3, suggesting that the main results are not driven by late-career, health-motivated job changes.

A separate concern is that estimated firm effects proxy geography or access rather than workplace conditions. In the Netherlands, job changes rarely coincide with home moves ($< 18\%$; Kronenberg and Carree, 2010), but I nonetheless add residential four-digit postcode fixed effects and, in a further specification, control for workplace-area healthcare access at PC4 (GP density, distance to pharmacies, and nearby hospitals). Table A14 shows that the estimated firm-effect dispersion changes little (0.025 versus 0.027), indicating that residential sorting and local access do not drive the findings.

Because variance components from the AKM decomposition are sensitive to limited-mobility bias, the baseline estimates use the leave-one-match-out correction of Kline et al. (2020). Table A15 shows that restricting to a highly connected subset of firms (at least 50 worker moves) reduces, but does not eliminate, plug-in bias and comes at substantial sample loss, while the KSS-corrected dispersion remains stable. Table A17 provides an additional validation by estimating firm effects in two random worker splits, and reporting a cross-split covariance (Panel B) that is very close to the full-sample KSS estimate of $\text{Var}(\psi_j)$ in Table 3, consistent with the baseline dispersion reflecting signal rather than estimation noise.

Finally, the baseline outcome uses concurrent annual expenditure, which is the natural margin for workplace influences in this setting: workers remain

See Appendix A7 for details. The mapping from majors to collar types uses a simple text-based classification implemented with a large language model, in line with recent evidence that such models match or exceed human accuracy in straightforward coding tasks (e.g. Gilardi et al., 2023; Chang et al., 2024).

tied to the employer during sickness absence for up to two years, and in the Grossman framework contemporaneous shifts in the health-depreciation rate map into contemporaneous expenditure flows. Consistent with this, Table A16 shows that firm-effect dispersion attenuates with the horizon when future expenditure is used as the dependent variable, reinforcing that the baseline estimates primarily capture contemporaneous firm effects on healthcare expenditure.

VI Discussion and Conclusion

The findings show that firms differ materially in the contribution they make to otherwise similar workers' healthcare expenditure. In the Dutch setting, where basic health insurance is universal, individual, and portable across jobs, these firm effects are unlikely to be driven by employer-specific insurance coverage or cost sharing. A one-standard-deviation difference in firm effects corresponds to a 17.8% difference in expected healthcare expenditure for the same worker, and exposure to firms with higher estimated healthcare-expenditure effects predicts higher subsequent disability and mortality. Together, these results show that firms make economically important contributions to workers' healthcare expenditure and that these contributions predict subsequent health.

The estimated firm effect should be interpreted as a composite measure of the workplace-related contribution to healthcare expenditure. It may reflect physical risks, psychosocial strain, work pace, task composition, safety practices, managerial quality, peer norms, and prevention or reintegration policies. Because job moves change several of these margins at once, and because detailed occupations, tasks, and workplace practices are not observed, the data do not allow a decomposition into separate mechanisms. The estimates instead capture the net effect of being employed at a particular firm on contemporaneous healthcare expenditure.

Several pieces of evidence point to workplace-related health risk as an important component of this net effect. Firms with higher healthcare-expenditure effects are concentrated in sectors with plausibly demanding physical or psychosocial environments, and firm effects are correlated with observable markers of demanding work environments, such as overtime. Workers who move to firms whose employees have higher average healthcare expenditure increase use of pain, anti-inflammatory, and muscle-relaxant medications, while comparable changes

are not visible for cardiometabolic or cancer-related medications. Firm effects also predict disability and mortality, including among workers displaced by firm closures. These patterns are difficult to reconcile with a pure utilisation or access interpretation.

Alternative explanations appear less central. Estimated firm-effect dispersion is stable after controlling for pay, residential location, and local healthcare access. It remains sizeable when the sample is restricted to workers who face near-zero marginal prices for additional care and when the outcome is GP expenditure, which is exempt from the deductible. These results suggest that prices, access, screening, and broad utilisation spillovers are unlikely to be the main drivers. Narrower peer effects in care-seeking may still matter, especially for musculoskeletal complaints, and peer-driven health behaviour could also play a role. But such channels are unlikely to fully explain the sharp changes in contemporaneous healthcare expenditure, and the downstream increases in disability and mortality.

The findings also sharpen the interpretation of compensating differentials. Firms with higher health-cost effects tend to pay higher wage premia, consistent with workers being partly compensated for less favourable health environments. The premium arises between industries, however; within detailed industries, firms with higher health-cost effects do not pay higher premia, so the firm-level component of health risk appears largely uncompensated in wages, subject to the measurement caveats in Section V. Yet even if wages adjust, the medical costs generated by these environments are largely financed collectively. In a system such as the Dutch one, this creates a fiscal externality: firms may not fully internalise the healthcare expenditure associated with the risks they generate. This does not mean that firms face no health-related costs. Dutch employers already bear substantial responsibilities through sickness pay, occupational-health obligations, reintegration requirements, and experience rating in disability insurance. However, basic healthcare expenditure remains financed collectively and firms do not directly internalise the healthcare costs they induce.

Several caveats are important. The outcome is healthcare expenditure under the statutory basic package, not health itself and not the full social cost of work-related health risks. Care financed through supplementary insurance is not observed. Some part of the estimated dispersion in firm effects may also reflect differences in

screening, referral, or care-seeking behaviour rather than underlying health.

Two directions for future research are especially important. The first is to identify and quantify the mechanisms through which firms affect health: physical strain, work intensity, schedule control, managerial practices, psychosocial stress, or peer behaviour. The second is to study whether stronger cost internalisation can reduce the dispersion in firm effects documented here. Broader experience rating or more employment-linked financing of healthcare could strengthen prevention incentives, but it could also create distortions, including selection against less healthy workers. Understanding this trade-off is central for policy design.

Overall, the results show that the firm should be treated as an important unit in the study of health disparities. Because workplace environments contribute to otherwise similar workers' healthcare expenditure and are linked to later health outcomes, policies focused only on individual behaviour or healthcare access will miss an important source of inequality.

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Appendices

A1 Institutional Background Details

This appendix provides additional institutional detail for the background discussion in Section II.

A1.I Dutch Healthcare and Insurance System

The Netherlands has a system of universal basic health insurance provided by private insurers under tight public regulation. All residents must purchase the basic package. Insurers must accept all applicants and charge community-rated premiums, while a national risk-equalization scheme compensates insurers for differences in enrollee risk. Financing is supplemented by tax revenue.⁴² Basic insurance is individual and portable: coverage does not change when workers change jobs. Employers may offer collective contracts, sometimes with premium discounts of up to about 10%, but these contracts do not alter the statutory content or cost-sharing rules of the basic package (Handel et al., 2020).

Cost sharing in the basic scheme is largely standardized through annual deductibles. In 2015, all adults faced a compulsory deductible of €375 and could choose an additional voluntary deductible of up to €500. The deductible applied to most specialist and hospital care and to many prescription drugs,⁴³ but not to GP care, maternity care, or several preventive services. More than 90% of adults, including almost all individuals with high expected healthcare expenditure, chose the minimum deductible (Handel et al., 2020). Switching between insurers was also infrequent: only 6.8% of individuals changed insurer in 2015 (Handel et al., 2020).

Access to care is organized around the general practitioner. Individuals are required to register with a GP, typically near their residence (Currie and Zwiers, 2025). GPs provide the first point of contact for non-emergency care and act as gatekeepers for almost all non-urgent specialist and hospital treatment. GP consultations are fully covered and exempt from the deductible. Pharmacies are

⁴²In 2015, the annual premium for a standard policy with the minimum deductible was around €1,200; low-income individuals received a healthcare benefit that offset part of the premium and deductible (Handel et al., 2020).

⁴³Prescription-drug cost sharing in the Dutch basic scheme is not identical across all medications, since some drugs can also be subject to drug-specific co-payments or reimbursement limits in addition to the general deductible.

widespread and usually located near GP practices. This organization makes basic access to primary care fairly uniform across workers and largely independent of the employer.

The content of the basic package is set by the government and is the same across insurers. It covers medically necessary GP care, specialist and hospital services, most prescription drugs, maternity and neonatal care, emergency transport, mental health services,⁴⁴ and selected medical aids. Routine adult dental care, eyeglasses and contact lenses, and many elective or alternative treatments are excluded. Physiotherapy is reimbursed only for chronic conditions and only after the first 20 sessions, so most short-term physiotherapy is either paid out of pocket or financed through supplementary insurance.⁴⁵

For this paper, the key implication is that basic insurance in the Netherlands is not attached to the employment relationship. Cross-firm differences in observed basic-package expenditure therefore arise in a setting with limited employer-driven variation in insurance generosity, provider networks, or statutory cost sharing.

A1.II Occupational Health, Screening, and Sickness Provisions

Dutch employers face extensive obligations in occupational health and sickness management. Under the Working Conditions Act (*Arbowet*), firms must contract a certified occupational health service or expert, maintain a written risk inventory and evaluation (*RI&E*) covering physical, chemical, ergonomic, and psychosocial risks, and implement a corresponding plan of action. Employers must also provide employees with access to a registered company doctor who supervises sickness absence and reintegration, and they must continue paying wages for up to two years during illness. When indicated by the *RI&E*, firms must offer a job-specific occupational health examination (*PAGO*), and they may also provide broader preventive medical examinations (*PMO*). Participation is voluntary except in a small set of legally defined high-risk or safety-critical jobs.

These obligations are reinforced by experience rating in disability-related

⁴⁴Although workplaces have expanded mental health and well-being services in recent years (Shreekumar and Vautrey, 2022), in the setting of this study most mental healthcare is still provided through the formal basic health insurance system (Prudon, 2025).

⁴⁵Vektis reports that 15.9% of insured persons in 2015 had no supplementary insurance, implying that about 84% held some form of supplementary insurance (Vektis, 2016, p. 23).

insurance. Employers pay an experience-rated contribution to the Return-to-Work Fund (Whk), which finances benefits under the Work and Income Act (WIA) and the Sickness Benefits Act (ZW). The premium depends on the firm's own inflow into sickness and disability benefits.⁴⁶ Dutch firms therefore face both legal duties and financial incentives to limit workplace-related health problems.

Compliance, however, is incomplete. Around 80% of firms report having a contract with an occupational health provider and roughly three quarters have a formal sickness-absence policy, but only about half maintain a written RI&E and fewer than 40% combine it with a concrete plan of action. When four core requirements are considered jointly—RI&E, an occupational health contract, in-company emergency response, and a designated prevention officer—only about one third of firms comply with all four ([Nederlandse Arbeidsinspectie, 2019](#)). Compliance is higher among large and high-risk firms than among small employers.

Preventive use of occupational health services is even narrower than formal coverage. Although most firms have access to a company doctor through an external provider, only about 7% report that employees used an open consultation hour in the previous year. Screening is similarly limited: PAGO/PMO is explicitly included in contracts for roughly one quarter of firms, but only about 4% report that an examination actually took place during the previous 12 months ([Nederlandse Arbeidsinspectie, 2019](#)). These programmes are concentrated in large firms and in sectors with well-identified exposures such as noise, chemicals, and heavy physical work. For most workers, day-to-day healthcare still runs through the ordinary GP-based system. Occupational doctors mainly provide work-related advice and reintegration support rather than medical treatment.

In short, the Dutch institutional setting combines substantial employer responsibility for prevention and sickness management with limited employer control over the content of ordinary medical care. That combination is useful for separating workplace health effects from insurance design.

⁴⁶Studies find that linking premiums to firm-specific disability inflow reduces DI entries among firms whose premiums actually adjust ([Koning, 2009](#)), and that removing experience rating raises DI inflow and lowers outflow for small employers ([De Groot and Koning, 2016](#)).

A1.III Workplace and Basic Healthcare Expenditure

Firms' contributions to basic healthcare expenditure can operate through two broad channels: differences in workers' ability to seek care and differences in the health consequences of the workplace itself.⁴⁷ The Dutch institutional setting is particularly informative because several features narrow the scope for the first channel. Out-of-pocket payments for basic care are capped by the annual deductible, more than 90% of adults choose the minimum deductible, and the marginal price of additional basic care is zero for high spenders once the deductible is exhausted. Existing evidence also finds limited income effects on healthcare expenditure (Miller et al., 2024; Cesarini et al., 2016). Time barriers also seem modest. Collective agreements and labour law provide paid time off for sickness and medical visits, employers must continue wage payments during illness for up to two years, and reintegration is formally supervised by occupational physicians. Consistent with this, self-reported unmet medical need in the Netherlands is among the lowest in the EU and shows little income gradient (e.g. OECD and European Observatory on Health Systems and Policies, 2019, 2021); lack of time is cited by less than 1% of respondents (Eurostat, 2023). Taken together, these features make the setting well suited to studying persistent firm effects on healthcare expenditure.

Workplace health conditions provide a natural set of mechanisms through which firms can have persistent effects on healthcare expenditure. One set of mechanisms runs through safety and physical working conditions: injury risks and ergonomic strain vary across firms and sectors and respond to regulation and enforcement (Hamermesh, 1999; Lee and Taylor, 2019; Johnson, 2020; Johansson et al., 2023). Despite substantial progress, occupational injuries and work-related musculoskeletal disorders remain major health hazards (Bhattacharya, 2014; van Ours, 2019). A second set concerns psychosocial stress: high workloads, long hours, job insecurity, and poor management practices are linked to stress, depression, burnout, and related somatic conditions (Hummels et al., 2025; Blackburn et al., 2023; Jolivet and Postel-Vinay, 2025). Third, workplaces differ in environmental exposures such as noise, dust, chemicals, heat, and air pollution, which are

⁴⁷Appendix A4 formalises this point in a Grossman-style framework (Grossman, 1972b), where individuals choose health investments given preferences, income and time constraints, and a workplace-specific health-depreciation rate.

established risk factors for respiratory, cardiovascular, renal, and cancer outcomes (e.g. [Collaborators et al., 2018](#); [Loomis et al., 2018](#); [Tran et al., 2022](#); [Gan et al., 2011](#)). Finally, firms may shape daily health behaviours through food provision, exercise incentives, and workplace norms around smoking, alcohol use, and physical activity ([Royer et al., 2015](#); [Jones et al., 2019](#); [Simonsen and Skipper, 2025](#)).

A2 Summary Statistics and Additional Results

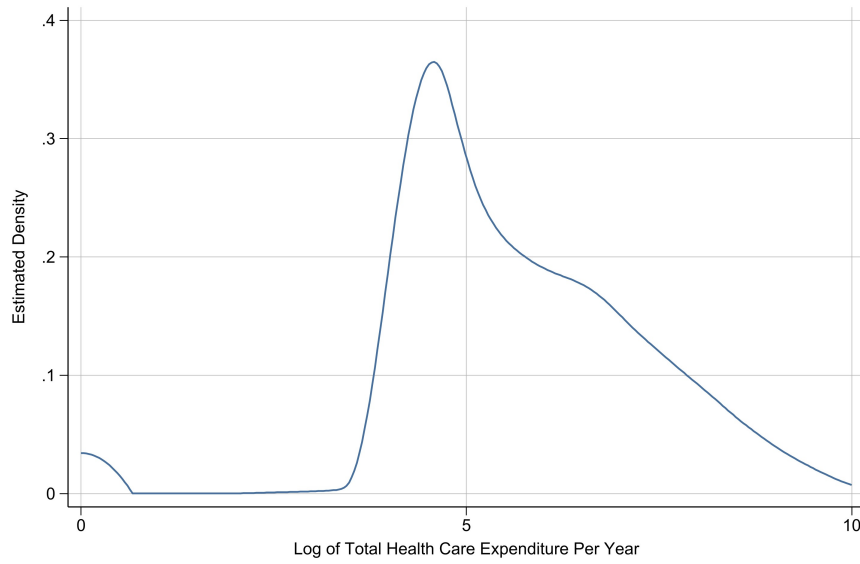


Figure A1: Log of total health expenditure per year for the connected set of employees (pooled male and female employees)

Statistic	Plug-In Estimates (Biased)
Log Health Exp. Var.: $var(h_i)$	3.01 (Ref.)
Firm FE Var.: $var(\psi_j)$	0.33 (10.9%)
Employee FE Var.: $var(\alpha_i)$	1.94 (64.4%)
Firm and Employee FE's Cov.: $cov(\alpha_i, \psi_j)$	-0.04
No. Movers	1,232,499
No. Firms	132,899
No. of Observations	17,300,199

Note: This table displays the variance and covariance components from Equation (2), using the naive plug-in estimates of the parameters estimated with the OLS method from Equation (2).

Table A1: The variance decomposition results using the fixed effects estimated in the TWFE model (2): Male workers

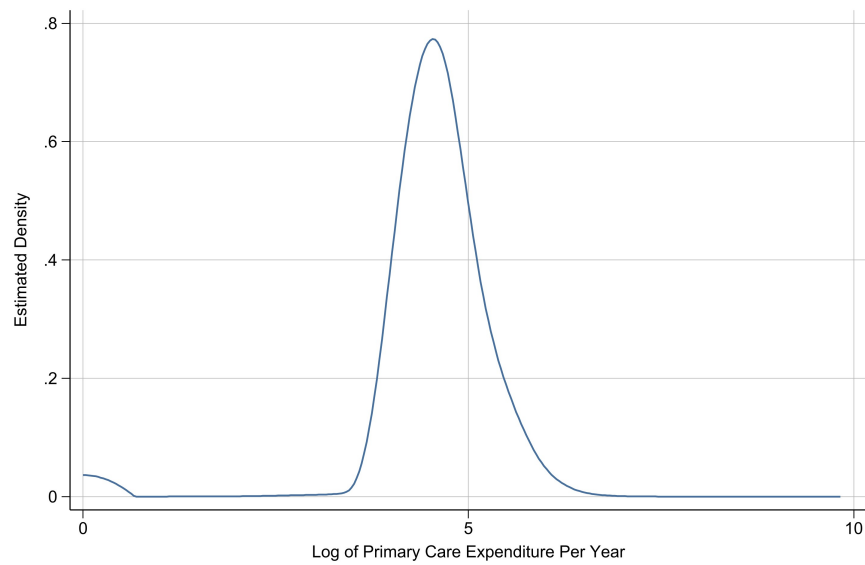


Figure A2: Log of primary health care expenditure per year for the connected set of employees (pooled male and female employees)

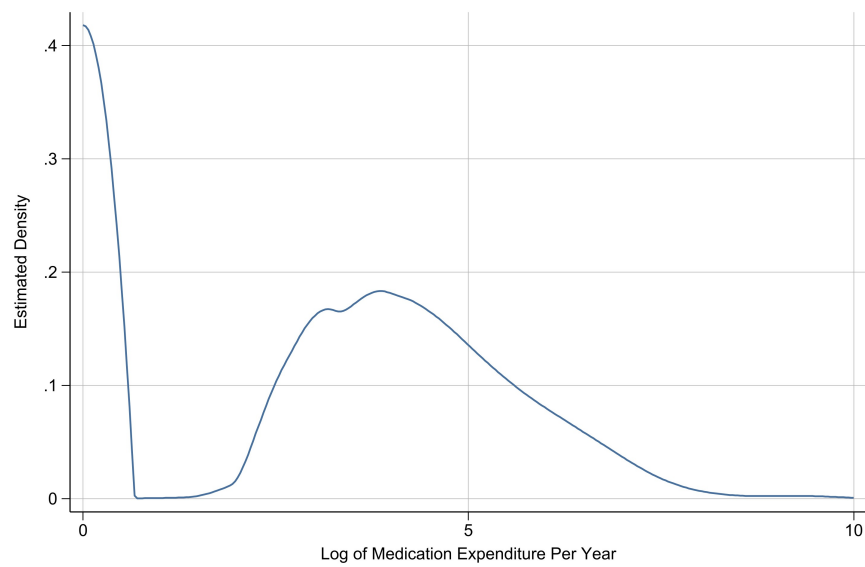


Figure A3: Log of medication costs per year for the connected set of employees (pooled male and female employees)

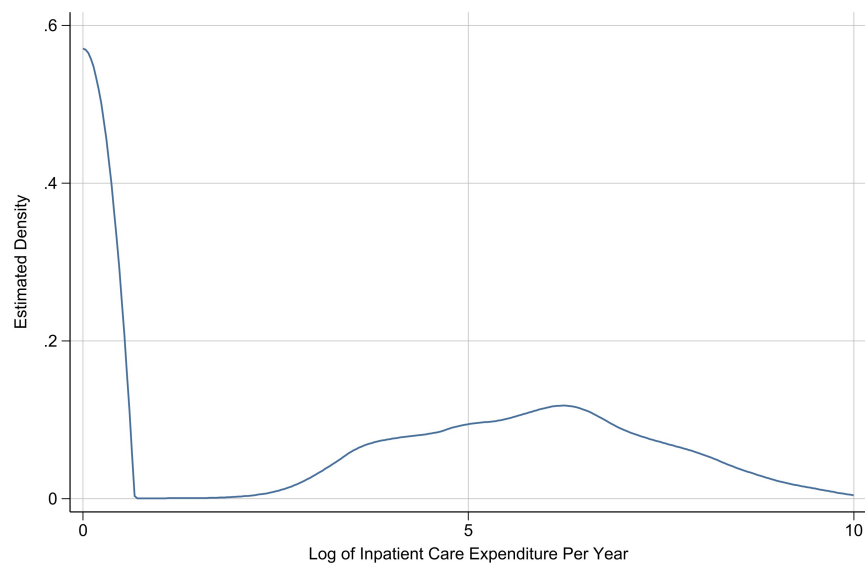


Figure A4: Log of inpatient care expenditure per year for the connected set of employees (pooled male and female employees)

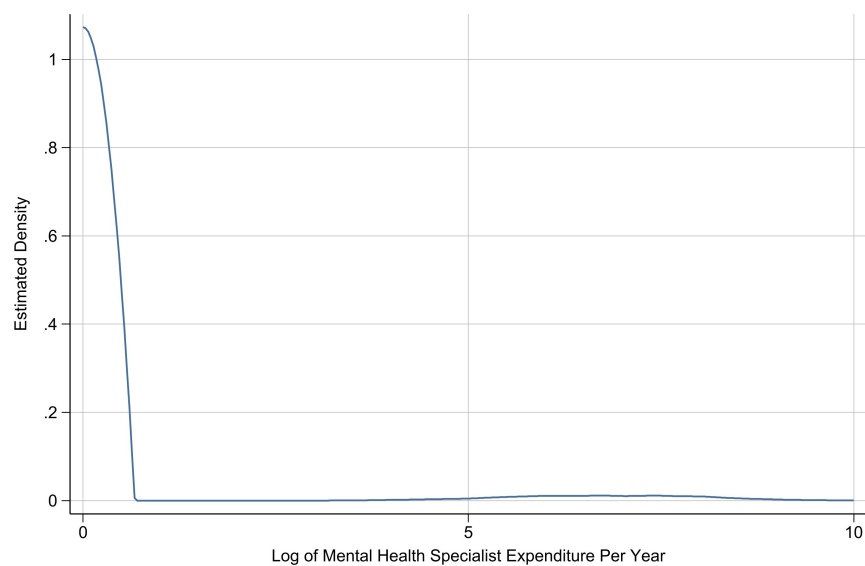
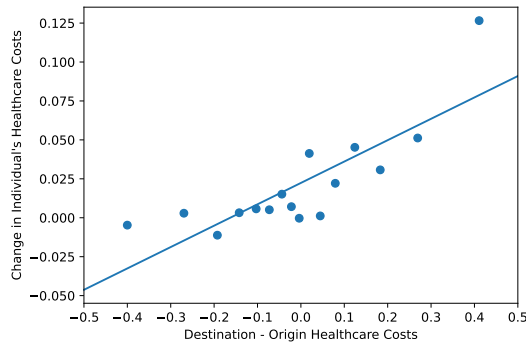
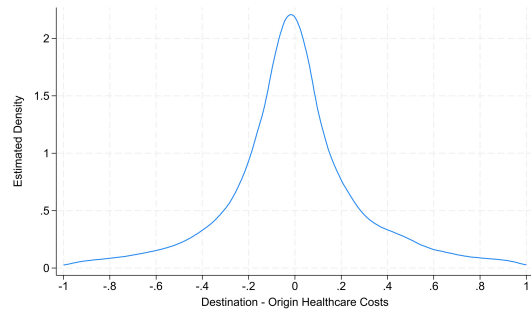


Figure A5: Log of mental health care expenditure (excluding primary care) per year for the connected set of employees (pooled male and female employees)

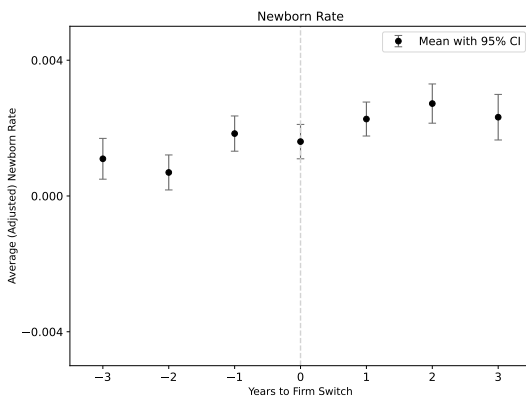


(a) Average Changes in Expenditure
(binscatter)

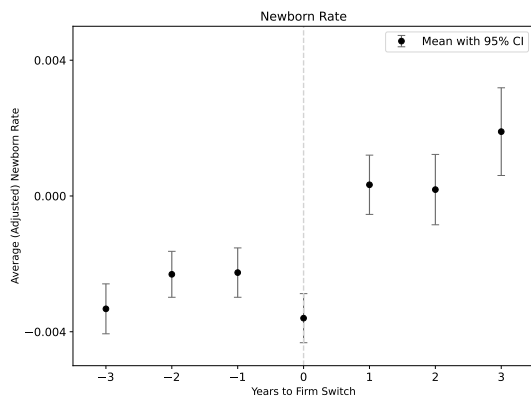


(b) Distribution of changes

Figure A6: Comparison of changes in workers' healthcare expenditure with changes in their colleagues' expenditure.



(a) Men



(b) Women

Note: The mean annual new birth rate is approximately 4% for both samples.

Figure A7: Average annual probability of having a new child around the time of a job switch (time = 0) for movers. Estimates are adjusted for age and education using a flexible age by education functional form using a linear regression.

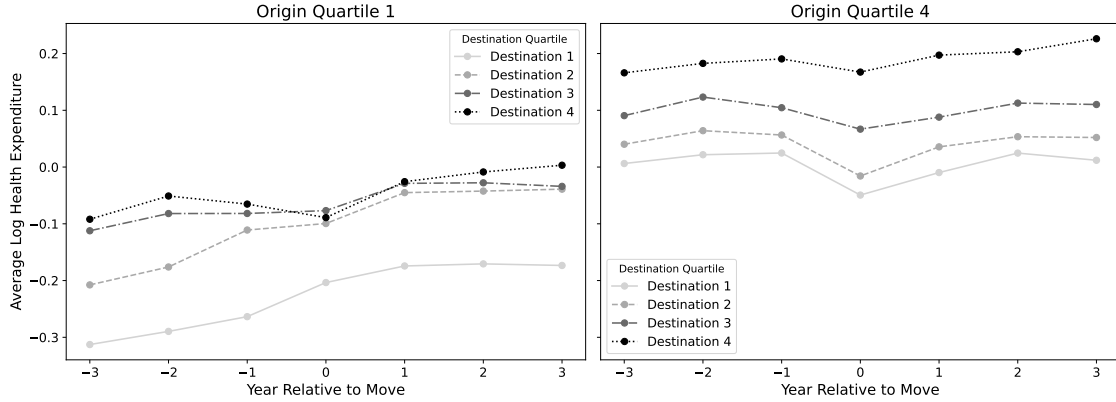


Figure A8: Mean age-adjusted log healthcare expenditures of movers by coworker expenditure quartile.

Note: Average age-adjusted logarithmic healthcare expenditures for employees who changed firms between 2009 and 2016, remaining at the origin firm for at least two years and at the destination firm for at least two years. Firms are grouped into quartiles by the average coworker health expenditure within the firm.

	Low group mean $\hat{h}_{it}$	High group mean $\hat{h}_{it}$	Difference (low – high)	Implied % difference in spending ^a
Household wealth 2009 (bottom vs top quintile)	0.147	-0.111	0.258	29.4
Total income 2009 (bottom vs top quintile)	0.153	-0.133	0.286	33.1
Education (primary vs master/doctor)	0.257	-0.140	0.398	48.8

^a The implied percentage difference is $[\exp(\text{Difference}) - 1] \times 100$.

Notes: Outcome is log annual basic-package healthcare spending residualised on age and calendar-year dummies, $\hat{h}_{it}$. Each cell reports the mean of $\hat{h}_{it}$ for the indicated group, averaged over person-year observations for full-time male employees aged 25–65 in the main analysis sample (2009–2016). Household wealth and total personal income are measured in 2009 and divided into five quintiles; the corresponding rows compare the bottom and top quintiles of these distributions. Education is measured as highest completed level, and the row compares workers with only primary education (“Basisonderwijs”) to those with a master’s degree or doctorate.

Table A2: Gradients in age- and year-adjusted log healthcare spending by SES

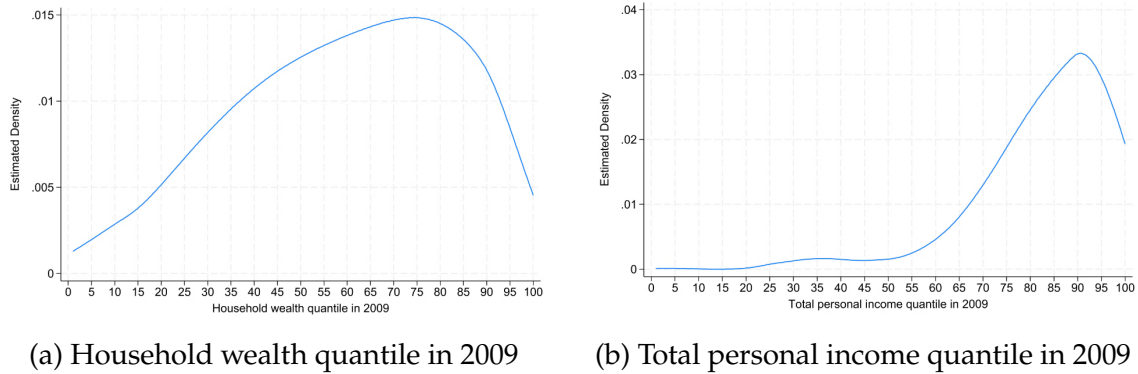


Figure A9: Location of the analysis sample in the national wealth and income distributions

Note: Each panel plots the kernel density of the position of the full-time male main sample in the national distribution of the corresponding SES measure, expressed as percentiles (0–100) of the Dutch population in 2009. For every individual in the main analysis sample, I use the Statistics Netherlands reported rank for 2009. The sample consists of full-time male employees aged 25–65 in the 2009–2016 analysis period.

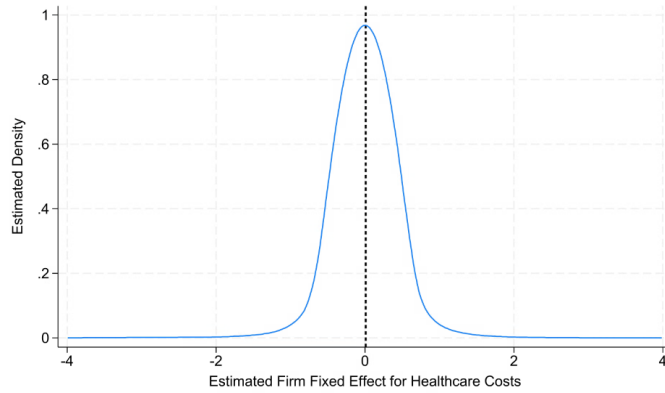


Figure A10: Distribution of estimated firm fixed effects

Note: Histogram of the demeaned estimated firm fixed effects $\hat{\psi}_j$ from the AKM specification in Equation (2) for all (individual-year weighted) firms in the connected set used in the main analysis. The outcome is log annual basic-package healthcare spending, $h_{it} = \log(c_{it} + 1)$, and the model includes worker and firm fixed effects as well as age–education profiles and calendar-year dummies. The solid line shows the median.

Statistic	KSS Estimates
Log Wage Var.: $\text{var}(w_i)$	0.176 (Ref.)
Firm FE Var.: $\text{var}(\psi_j^w)$	0.017 (9.48 %)
Employee FE Var.: $\text{var}(\alpha_i^w)$	0.271 (154.12 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i^w, \psi_j^w)$	−0.001
Correlation $\rho(\alpha_i^w, \psi_j^w)$	−0.020
Explained share: workers & firms	1.621
No. Movers	1,225,799
No. Firms	132,820
No. of Observations	16,976,071
Average Log Wage: $\text{mean}(w_i)$	3.141

Notes: Components estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). The outcome is log hourly wage. Hourly wages are constructed from gross annual contract earnings, contractual full-time equivalent (FTE), and contract length, assuming that a full-time job corresponds to 52 weeks of 40 hours per week, as in the data construction in Section III. The sample consists of full-time male employees in the leave-one-out connected set of firms between 2009 and 2016. Shares in parentheses are relative to $\text{var}(w_i)$.

Table A3: Variance Decomposition of Log Hourly Wages

Panel A: KSS Variance Components	
Statistic	KSS Estimates
Log Health Exp. Var.: $\text{var}(h_i)$	2.617 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.027 (1.03 %)
Employee FE Var.: $\text{var}(\alpha_i)$	1.515 (57.89 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	−0.004
Correlation $\rho(\alpha_i, \psi_j)$	−0.018
Explained share: workers & firms	0.586
No. Movers	1,225,799
No. Firms	132,820
No. of Observations	16,976,071
Average Health Exp.: $\text{mean}(h_i)$	5.608
Panel B: Regression of Firm Effects with Pay Control on Baseline Firm Effects	
Dependent variable	$\hat{\psi}_j^{\text{HC, pay}}$
Coefficient on baseline firm FE (worker level) $\hat{\psi}_j$	1.000
No. of workers	16,976,071

Notes: Panel A reports components estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). The outcome is log annual basic-package healthcare expenditure per calendar year. The regression includes worker and firm fixed effects, age-by-education profiles, calendar-year dummies, and an additional control for workers’ annual gross contract payment (yearly labour income). Shares in parentheses are relative to $\text{var}(h_i)$. Panel B reports an OLS regression of the firm fixed effects from the healthcare model with the contract-payment control, $\hat{\psi}_j^{\text{HC, pay}}$, on the baseline healthcare firm fixed effects without that control, $\hat{\psi}_j$. The slope is close to one and the intercept is close to zero, indicating that adding the pay control leaves the pattern of estimated firm health effects almost unchanged.

Table A4: Variance Decomposition of Log Healthcare Expenditure (Controlling for Workers’ Annual Contract Payments)

	High baseline costs ($\hat{\alpha}_i$ above median)	Low baseline costs ($\hat{\alpha}_i$ below median)
<i>Panel A. KSS variance components</i>		
Log Health Exp. Var.: $\text{var}(h_i)$	2.101 (Ref.)	1.745 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.022 (1.03 %)	0.035 (1.98 %)
Employee FE Var.: $\text{var}(\alpha_i)$	0.550 (26.19 %)	0.563 (32.25 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	0.032	0.032
Correlation $\rho(\alpha_i, \psi_j)$	0.297	0.228
Explained share: workers & firms	0.303	0.379
No. Movers	551,201	606,960
No. Firms	80,230	89,384
No. of Observations	8,035,638	8,110,825
Mean log Health Exp.: $\text{mean}(h_i)$	6.447	4.775
<i>Panel B. Descriptives and cross-sample link</i>		
Mean annual basic-package costs (€) ^a	2,011	286.52
No. obs. used for mean costs ^a	8,488,038	8,488,033
Bias-corrected slope (worker level), firm FE (low baseline) on firm FE (high baseline) ^b	0.602	
No. worker–year obs. in slope regression ^b	7,635,508	

^a Mean annual basic-package costs are computed directly from raw annual spending c_{it} (in euros) in the two samples, split by whether the estimated individual healthcare fixed effect $\hat{\alpha}_i$ from the baseline AKM lies above or below its median (median split defined over all connected-set worker–year observations).

^b Slope from an OLS regression of the firm fixed effects estimated in the low-baseline-costs subsample on the firm fixed effects estimated in the high-baseline-costs subsample, using worker–year observations in the overlap where both firm effects are defined. The reported slope is bias-corrected by rescaling the attenuated OLS coefficient using the ratio of KSS-corrected to plug-in firm-effect variances following [Kline et al. \(2020\)](#). Because KSS is implemented on the leave-one-out connected set *within each subsample*, the two subsamples need not form an exact partition of the baseline connected set, and some firms (and their worker–year observations) appear in only one subsample and therefore drop out of the overlap regression.

Notes: Panel A components are estimated via leave-one-match-out KSS correction ([Kline et al., 2020](#)) using random projections (50 simulations). The outcome is log annual basic-package healthcare expenditure per calendar year, $h_{it} = \log(c_{it} + 1)$. Shares in parentheses are relative to $\text{var}(h_i)$ in the respective subsample.

Table A5: Variance Decomposition of Log Healthcare Expenditure by Baseline Individual Health-Cost Type

	High-wage workers (above-median hourly wage)	Low-wage workers (below-median hourly wage)
<i>Panel A. KSS variance components</i>		
Log Health Exp. Var.: $\text{var}(h_i)$	2.543 (Ref.)	2.706 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.022 (0.88 %)	0.039 (1.43 %)
Employee FE Var.: $\text{var}(\alpha_i)$	1.198 (47.08 %)	1.630 (60.22 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	−0.006	−0.005
Explained share: workers & firms	0.475	0.613
No. Movers	547,051	631,701
No. Firms	58,556	103,831
No. of Observations	8,061,912	8,278,053
Mean log Health Exp.: $\text{mean}(h_i)$	5.567	5.653
<i>Panel B. Descriptives and cross-sample link</i>		
Mean hourly wage (€) ^a	33.13	16.39
Mean annual contract payment (€) ^a	67,783	33,306
No. obs. used for wage descriptives ^a	8,488,041	8,488,030
Bias-corrected slope (worker level), firm FE (low wage) on firm FE (high wage) ^b		0.622
No. worker–year obs. in slope regression ^b		6,644,986

^a Wage groups are defined using an estimated hourly wage (in euros) computed from annual contract payment and work intensity (as in Section III). The median hourly wage in the connected-set sample is 20.900,00; “high wage” corresponds to observations above this threshold. Panel B means are computed directly from the raw wage variables in the connected-set person–year sample and are descriptive.

^b Slope from an OLS regression of the firm fixed effects estimated in the low-wage subsample on the firm fixed effects estimated in the high-wage subsample, using worker–year observations in the overlap where both firm effects are defined. The reported slope is bias-corrected by rescaling the attenuated OLS coefficient using the ratio of KSS-corrected to plug-in firm-effect variances following Kline et al. (2020). Because KSS is implemented on the leave-one-out connected set *within each wage subsample*, the two subsamples need not form an exact partition of the baseline connected set, and some firms (and their worker–year observations) appear in only one subsample and therefore drop out of the overlap regression.

Notes: Panel A components are estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). The outcome is log annual basic-package healthcare expenditure per calendar year, $h_{it} = \log(c_{it} + 1)$. Shares in parentheses are relative to $\text{var}(h_i)$ in the respective subsample.

Table A6: Variance Decomposition of Log Healthcare Expenditure by Wage Group

	Young workers (age ≤ 45)	Old workers (age > 45)
<i>Panel A. KSS variance components</i>		
Log Health Exp. Var.: $\text{var}(h_i)$	2.390 (Ref.)	2.680 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.029 (1.23 %)	0.014 (0.54 %)
Employee FE Var.: $\text{var}(\alpha_i)$	1.369 (57.28 %)	1.649 (61.52 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	0.005	−0.001
Explained share: workers & firms	0.590	0.620
No. Movers	742,335	409,353
No. Firms	108,472	49,383
No. of Observations	8,653,958	7,154,491
Mean log Health Exp.: $\text{mean}(h_i)$	5.317	5.944
<i>Panel B. Cross-sample link</i>		
Bias-corrected slope (worker level), firm FE (young) on firm FE (old) ^a	0.846	
No. worker–year obs. in slope regression ^a	7,236,057	

^a Slope from an OLS regression of the firm fixed effects estimated in the young subsample on the firm fixed effects estimated in the old subsample, using worker–year observations in the overlap where both firm effects are defined. The reported slope is bias-corrected by rescaling the attenuated OLS coefficient using the reliability ratio for the right-hand-side firm effect, $\text{Var}_{\text{KSS}}(\psi)/\text{Var}_{\text{PI}}(\hat{\psi})$, following [Kline et al. \(2020\)](#) (equivalently, $\hat{\beta}_{\text{BC}} = \hat{\beta}_{\text{OLS}} \times \text{Var}_{\text{PI}}(\hat{\psi}_{\text{old}})/\text{Var}_{\text{KSS}}(\psi_{\text{old}})$). Because KSS is implemented on the leave-one-out connected set *within each age subsample*, the two subsamples need not form an exact partition of the baseline connected set; some firms appear in only one subsample and drop out of the overlap regression.

Notes: Panel A components are estimated via leave-one-match-out KSS correction ([Kline et al., 2020](#)) using random projections (50 simulations). The outcome is log annual basic-package healthcare expenditure per calendar year, $h_{it} = \log(c_{it} + 1)$. Shares in parentheses are relative to $\text{var}(h_i)$ in the respective subsample.

Table A7: Variance Decomposition of Log Healthcare Expenditure
by Age Group

	White-collar workers (education proxy)	Blue-collar workers (education proxy)
<i>Panel A. KSS variance components</i>		
Log Health Exp. Var.: $\text{var}(h_i)$	2.142 (Ref.)	2.286 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.020 (0.92 %)	0.032 (1.38 %)
Employee FE Var.: $\text{var}(\alpha_i)$	1.071 (50.01 %)	1.143 (49.99 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	−0.005	−0.012
Explained share: workers & firms	0.504	0.503
No. Movers	319,969	158,616
No. Firms	49,485	40,897
No. of Observations	3,719,229	1,903,635
Mean log Health Exp.: $\text{mean}(h_i)$	5.443	5.638
<i>Panel B. Cross-sample link</i>		
Bias-corrected slope (worker level), firm FE (white collar) on firm FE (blue collar) ^b		1.063
No. worker–year obs. in slope regression ^b		1,598,774

^a Blue- and white-collar status is proxied using education (see Appendix A7 for the construction of the collar proxy used in the paper).

^b Slope from an OLS regression of the firm fixed effects estimated in the white-collar subsample on the firm fixed effects estimated in the blue-collar subsample, using worker–year observations in the overlap where both firm effects are defined. The reported slope is bias-corrected by rescaling the attenuated OLS coefficient using the right-hand-side reliability ratio $\text{Var}_{\text{KSS}}(\psi)/\text{Var}_{\text{PI}}(\hat{\psi})$ following Kline et al. (2020), i.e. $\hat{\beta}_{\text{BC}} = \hat{\beta}_{\text{OLS}} \times \text{Var}_{\text{PI}}(\hat{\psi}_{\text{blue}})/\text{Var}_{\text{KSS}}(\psi_{\text{blue}})$.

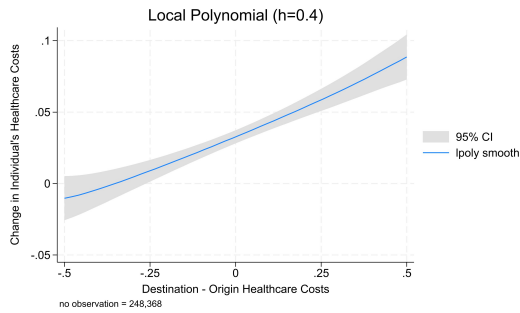
Notes: Panel A components are estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). The outcome is log annual basic-package healthcare expenditure per calendar year, $h_{it} = \log(c_{it} + 1)$. Shares in parentheses are relative to $\text{var}(h_i)$ in the respective subsample.

Table A8: Variance Decomposition of Log Healthcare Expenditure by Collar Type (proxied by education)

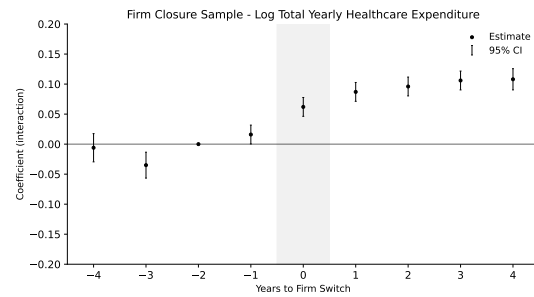
Statistic	Primary care (GP)	Medication	Hospital (Inpatient + Specialist)	Mental health
Log Health Exp. Var.: $\text{var}(h_i)$	0.655	5.689	9.811	1.632
Firm FE Var.: $\text{var}(\psi_j)$	0.018	0.010	0.024	0.003
Employee FE Var.: $\text{var}(\alpha_i)$	0.521	3.209	2.911	0.371
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	−0.003	−0.001	−0.004	−0.002
Correlation $\rho(\alpha_i, \psi_j)$	−0.027	−0.006	−0.017	−0.061
Explained share: worker & firm	0.815	0.565	0.298	0.227
No. Movers	1,225,799	1,225,799	1,225,799	1,225,799
No. Firms	132,820	132,820	132,820	132,820
No. of Observations	16,976,071	16,976,071	16,976,071	16,976,071
Average Health Exp.: $\text{mean}(h_i)$	4.521	2.637	2.732	0.240

Notes: All columns report leave-one-match-out KSS estimates (Kline et al., 2020). Outcomes are the log of subcategory-specific healthcare expenditure: primary care (GP), medication, hospital (inpatient + specialist), and specialist mental health. The sample is the full connected set.

Table A9: Variance Decomposition of Log Healthcare Expenditure (KSS) by Cost Category



(a) Average change vs. destination exposure gap



(b) Event study around closure and reemployment

Figure A11: Workers' healthcare spending around firm closures and reemployment

Notes. Sample of full-time male employees aged 25–65 whose employer exits during the panel and who are observed in consecutive years from 2009 and around the closure. Healthcare outcomes are log total basic-package expenditures, residualised on age and calendar-year dummies. The closure year is the last year in which the origin employer identifier is observed. In panel (a), each point plots the average change in individual annual residualised spending from the pre-closure year to the post-reemployment year against the destination exposure gap, defined as the difference in leave-*i*-out mean residualised spending between the destination and origin employers. The line shows a non-parametric fit, where the intercept captures the common post-displacement increase and the slope measures pass-through associated with the destination exposure gap. In panel (b), the markers show coefficients from an event-study specification that interacts event-time indicators with the destination exposure gap, with event time measured relative to the closure year and $t = -2$ as the reference year.

Closure Sample (Worker-year observations) N = 1,550,889	
Variable	Mean (SD)
Age	46.03 (9.43)
Dutch background (dummy)	0.85 (0.36)
First-generation migrant (dummy)	0.08 (0.27)
Second-generation migrant (dummy)	0.07 (0.25)
Education: Elementary (dummy)	0.02 (0.13)
Education: Lower sec./MBO1 (dummy)	0.05 (0.21)
Education: Upper sec./MBO (dummy)	0.19 (0.40)
Education: Bachelor (HBO/WO) (dummy)	0.18 (0.38)
Education: Master/Doctorate (dummy)	0.11 (0.31)
Education: Missing (dummy)	0.46 (0.50)
Total # children	1.54 (1.20)
New child (dummy)	0.04 (0.19)
Gross payment of contract (€)	55,101 (40,796)
Mover (dummy)	1.00 (0.00)
Full-time factor (FTE)	1.00 (0.01)
Estimated hourly wage (€) ^a	26.84 (19.78)
Business size (employees)	1,012 (815.15)
Total health care costs (€)	1,159 (4,574)
Mental health (specialist) (€)	75.13 (1,188)
Hospital costs (€)	700.58 (3,867)
GP costs (€)	110.45 (65.88)
Number of Observations	1,550,889

^a Hourly wage computed from gross contract payment assuming a full FTE works 52 weeks × 40 hours per week.

Notes: Firm-closure sample (worker-year observations). Variables are defined as in Table 1. All observations are movers by construction (Mover dummy = 1). Business size is observed for 1,544,519 worker-year observations (all other variables are observed for 1,550,889).

Table A10: Summary Statistics: Firm-Closure Sample

	(1)	(2)	(3)
	<i>Outcome measured pre-move</i>		
Destination firm health-cost effect (sample-split)	0.0068 (0.0092)	-0.3305*** (0.0132)	-0.3879*** (0.0201)
Mean of dep. var.	0.2054	0.2054	0.1960
<i>N</i>	209,235	209,235	206,945
<i>R</i> ²	0.084	0.089	0.146
Baseline controls & pre-move postal-code FEs	Yes	Yes	Yes
Origin firm-effect level (control)	No	Yes	No
Origin-firm fixed effects	No	No	Yes

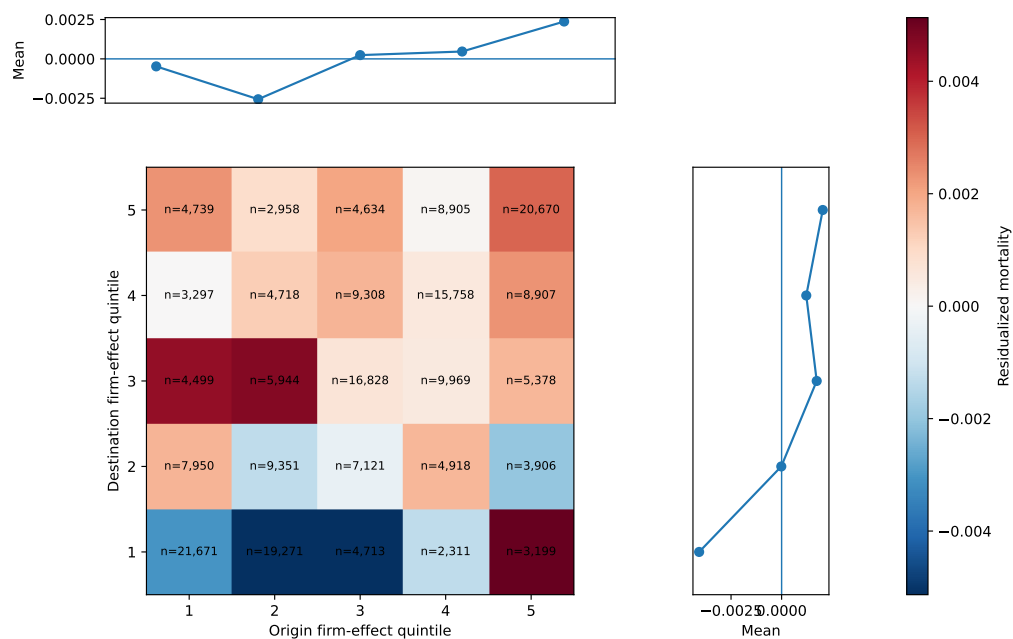
Notes: Sample of displaced male workers following firm closures, restricting destination firms to those above the 25th percentile of firm size (at least 75 employees). The outcome is average pre-move basic-package healthcare spending. Covariates are the same as in Table 5. Column (2) additionally controls for the origin firm-effect level; column (3) includes origin-firm fixed effects. Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A11: Firm Closures: Pre-Move Healthcare Spending

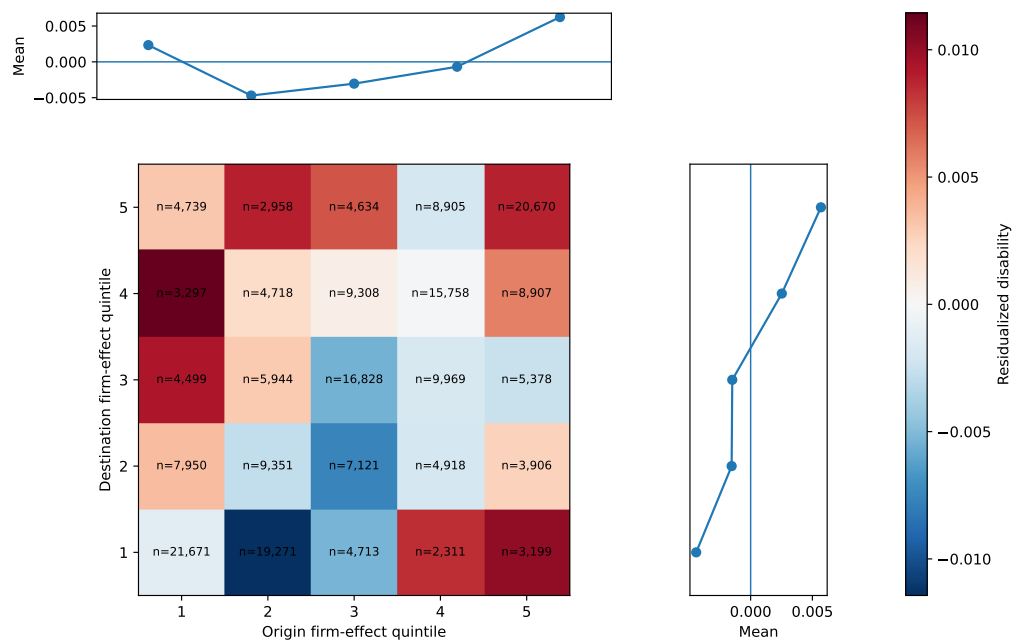
	(1)	(2)	(3)
<i>Outcome measured in 2019</i>			
Panel A: Mortality (Accidental/Injuries)			
Destination firm health-cost effect (sample-split)	0.000021 (0.000128)	0.000171 (0.000184)	-0.000170 (0.000282)
Mean of dep. var.		0.00034	
N	209,235	209,235	206,945
R ²	0.020	0.020	0.072
Panel B: Mortality (Cancers)			
Destination firm health-cost effect (sample-split)	0.001772*** (0.000570)	0.002752*** (0.000820)	0.002729** (0.001267)
Mean of dep. var.		0.00687	
N	209,235	209,235	206,945
R ²	0.037	0.037	0.069
Panel C: Mortality (Cardiometabolic)			
Destination firm health-cost effect (sample-split)	0.000341 (0.000366)	0.000774 (0.000527)	0.000570 (0.000815)
Mean of dep. var.		0.00278	
N	209,235	209,235	206,945
R ²	0.024	0.024	0.057
Panel D: Mortality (Chronic Respiratory)			
Destination firm health-cost effect (sample-split)	0.000086 (0.000084)	0.000131 (0.000121)	0.000091 (0.000189)
Mean of dep. var.		0.00015	
N	209,235	209,235	206,945
R ²	0.028	0.028	0.046
Panel E: Mortality (Suicide/Substance)			
Destination firm health-cost effect (sample-split)	0.000050 (0.000217)	-0.000458 (0.000312)	-0.000906* (0.000477)
Mean of dep. var.		0.00096	
N	209,235	209,235	206,945
R ²	0.019	0.019	0.061
Panel F: Mortality (Other)			
Destination firm health-cost effect (sample-split)	0.000990*** (0.000313)	0.001377*** (0.000451)	0.001039 (0.000698)
Mean of dep. var.		0.00203	
N	209,235	209,235	206,945
R ²	0.019	0.019	0.052
Baseline controls & pre-move postal-code FEs	Yes	Yes	Yes
Origin firm-effect level (control)	No	Yes	No
Origin-firm fixed effects	No	No	Yes

Notes: Sample of displaced male workers following firm closures, restricting destination firms to those above the 25th percentile of firm size (at least 75 employees). The regressor is the sample-split destination-firm health-cost effect. Column (2) additionally controls for the sample-split origin firm-effect level; column (3) includes origin-firm fixed effects. Baseline controls include demographics and pre-move postal-code fixed effects (as in Table 5). Cause-specific outcomes are constructed from the underlying cause-of-death ICD-10 codes recorded in the mortality register. Standard errors in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A12: Firm Closures: Destination-Firm Health-Cost Effect and Cause-Specific Mortality in 2019



(a) Mortality in 2019



(b) Disability in 2019

Figure A12: Long-run outcomes by origin×destination firm-effect quintiles after firm closures

Note: Sample and residualization are the same as in Figure 5. The sample-split origin and destination firm health-cost effects are binned into quintiles (1 = lowest, 5 = highest). Cells show binned means of residualized outcomes, and annotations report the number of observations in each origin×destination cell.

Statistic	KSS Estimates
Log Health Exp. Var.: $\text{var}(h_i)$	2.609 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.013 (0.49 %)
Employee FE Var.: $\text{var}(\alpha_i)$	1.402 (53.74 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	0.006
Correlation $\rho(\alpha_i, \psi_j)$	0.045
Explained share: workers & firms	0.547
No. Movers	1,113,768
No. Firms	117,247
No. of Observations	16,737,318
Average Health Exp.: $\text{mean}(h_i)$	5.611

Notes: Components estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). Outcome is log healthcare expenditure per calendar year. The sample is restricted to full-time employees in firms with a non-missing sector code in the business registry, 2009–2016. Relative to the full connected sample in Table 3, the variance of firm effects is smaller in this restricted set, so firm effects account for a slightly smaller share of $\text{var}(h_i)$. Shares in parentheses are relative to $\text{var}(h_i)$.

Table A13: Variance Decomposition of Log Healthcare Expenditure: Registry-Sector Sample

Statistic	PC4 FE (Residence)	PC4 FE + Workplace Access
Log Health Exp. Var.: $\text{var}(h_i)$	2.393 (Ref.)	2.386 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.025 (1.03 %)	0.025 (1.03 %)
Employee FE Var.: $\text{var}(\alpha_i)$	1.360 (56.85 %)	1.368 (57.32 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	−0.006	−0.007
Correlation $\rho(\alpha_i, \psi_j)$	−0.031	−0.037
Explained share: worker, firm & edu-age	0.574	0.578
No. Movers	1,212,637	1,173,299
No. Firms	131,992	129,198
No. of Observations	16,806,677	16,263,162
Average Health Exp.: $\text{mean}(h_i)$	5.652	5.653

Notes: Components estimated via leave-one-match-out KSS correction (Kline et al., 2020) using random projections (50 simulations). Outcome is log healthcare expenditure per calendar year. The left column includes four-digit postcode (PC4) fixed effects for workers’ residential location; the right column additionally controls for PC4-level workplace-area healthcare access (number of GPs, distance to nearest pharmacy, number of nearby hospitals). Shares in parentheses are relative to $\text{var}(h_i)$.

Table A14: Variance Decomposition of Log Healthcare Expenditure (KSS, with Residential PC4 FEs and Workplace Access Controls)

Statistic	Plug-in (Full Sample)	Plug-in (≥ 50 -Move Firms)	KSS (≥ 50 -Move Firms)
Log Health Exp. Var.: $\text{var}(h_i)$	2.617 (Ref.)	2.675 (Ref.)	2.675 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.113 (4.33 %)	0.049 (1.84 %)	0.030 (1.13 %)
Employee FE Var.: $\text{var}(\alpha_i)$	1.839 (70.28 %)	1.790 (66.91 %)	1.520 (56.82 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	−0.077	−0.026	−0.008
Correlation $\rho(\alpha_i, \psi_j)$	−0.169	−0.086	−0.039
Explained share: worker, firm & edu–age	0.687	0.668	0.573
No. Movers	1,225,799	723,530	723,530
No. Firms	132,820	10,643	10,643
No. of Observations	16,976,071	11,295,814	11,295,814
Average Health Exp.: $\text{mean}(h_i)$	5.608	5.631	5.631

Notes: Plug-in columns report naive AKM quadratic components. The KSS column applies the leave-one-match-out bias correction of [Kline et al. \(2020\)](#). Outcome is log healthcare expenditure per calendar year. The ≥ 50 -move sample restricts to firms with at least 50 observed worker moves. Percent shares in parentheses are relative to $\text{var}(h_i)$.

Table A15: Variance Decomposition of Log Healthcare Expenditure: Plug-in vs. KSS, Full Sample vs. ≥ 50 -Move Firms

Statistic	1-year Ahead	2-year Ahead	3-year Ahead	4-year Ahead	5-year MA
Log Health Exp. Var.: $\text{var}(h_i)$	2.645 (Ref.)	2.723 (Ref.)	2.885 (Ref.)	3.044 (Ref.)	1.763 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.023	0.009	0.004	0.004	0.009
Employee FE Var.: $\text{var}(\alpha_i)$	1.490	1.544	1.662	1.723	1.610
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	−0.003	−0.001	−0.001	−0.003	0.000
Correlation $\rho(\alpha_i, \psi_j)$	−0.018	−0.005	−0.007	−0.039	−0.003
Explained share: worker, firm & edu–age	0.569	0.570	0.577	0.565	0.918
No. Movers	1,225,799	1,225,799	1,225,799	1,225,799	1,225,799
No. Firms	132,820	132,820	132,820	132,820	132,820
No. of Observations	16,976,071	16,976,071	16,976,071	16,976,071	16,976,071
Average Health Exp.: $\text{mean}(h_i)$	5.669	5.706	5.722	5.739	5.689

Notes: All columns report leave-one-match-out KSS estimates ([Kline et al., 2020](#)). Outcomes are the log of healthcare expenditure measured at different horizons relative to the baseline year: one-through four-year ahead, and a five-year moving average (current plus next four years). The sample is the full connected set.

Table A16: Variance Decomposition of Log Healthcare Expenditure (KSS) Across Horizons

A3 Mover Evidence and Firm-Effect Dispersion

This appendix clarifies what the mover evidence measures. The event study asks how much of the origin–destination gap in firm-average healthcare expenditure passes through after a move. The main estimates instead recover firm effects and quantify both their dispersion across the workforce and their contribution to individual-level expenditure variation. These are different empirical objects; the formal results are developed in [Moghani \(2026\)](#).

Consider a two-period model with worker and firm fixed effects,

$$Y_{it} = \alpha_i + \psi_{j(i,t)} + \varepsilon_{it}, \quad (3)$$

where α_i is an individual effect and ψ_j is a firm effect. Let $\bar{\alpha}_j$ denote the average individual effect among workers at firm j , so that the firm’s average outcome is

$$\bar{Y}_j = \psi_j + \bar{\alpha}_j.$$

For a mover from origin $o(i)$ to destination $d(i)$, define $\Delta\psi_i = \psi_{d(i)} - \psi_{o(i)}$ and $\Delta\bar{\alpha}_i = \bar{\alpha}_{d(i)} - \bar{\alpha}_{o(i)}$. With sufficiently large firms, the origin–destination gap in average outcomes converges to

$$\Delta\bar{Y}_i = \Delta\psi_i + \Delta\bar{\alpha}_i.$$

Under the usual exogenous-mobility condition, the mover coefficient therefore converges to

$$\theta_M = \frac{\text{Cov}_M(\Delta\bar{Y}_i, \Delta\psi_i)}{\text{Var}_M(\Delta\bar{Y}_i)}. \quad (4)$$

Equation (4) shows that the mover coefficient is a transition parameter. It depends on which firms movers leave and enter, how far those moves travel in the firm-effect distribution, and how changes in firm effects covary with changes in worker composition along those routes. Economies with the same cross-sectional dispersion of firm effects and the same sorting of workers across firms can therefore produce different mover coefficients when their mobility patterns differ.

A cross-sectional interpretation is available when moves reproduce the cross-sectional variation in firm effects and worker composition in the same proportions.

Moghani (2026) calls this condition *directional isotropy*. In compact form, for some scalar $\kappa > 0$,

$$\text{Var}_M \begin{pmatrix} \Delta\psi_i \\ \Delta\bar{\alpha}_i \end{pmatrix} = \kappa \text{Var}_M \begin{pmatrix} \psi_j \\ \bar{\alpha}_j \end{pmatrix}, \quad (5)$$

where cross-sectional moments weight firms by their incidence among movers. Under this condition,

$$\theta_M = \frac{\text{Cov}_M(\psi_j, \bar{Y}_j)}{\text{Var}_M(\bar{Y}_j)}.$$

Crucially, this equality concerns only the cross-sectional relationship between firm effects and firm-average outcomes. Both ψ_j and $\bar{Y}_j$ are firm-level objects, and the moments are computed across firms visited by movers. It does not recover the individual-level firm share, $\text{Var}(\psi_{j(i,t)}) / \text{Var}(Y_{it})$. That denominator also contains variation among workers within firms and transitory expenditure shocks, and its workforce weights generally differ from the mover-incidence weights above. Directional isotropy aligns observed transitions with cross-sectional variation in firm effects and worker composition; it does not connect variation in firm averages to variation in individual healthcare expenditure.

Even at the firm-average level, the mover coefficient is not generically a variance share or a standard-deviation share. It lies midway between the share of cross-firm variance attributable to firm effects and the share of cross-firm variance that would disappear if firm effects were equalized. If firm effects are orthogonal to worker composition and mobility is directionally isotropic, it equals $\text{Var}_M(\psi_j) / \text{Var}_M(\bar{Y}_j)$. If firm composition is an affine, residual-free function of the firm effect with a non-negative slope, it equals $\text{sd}_M(\psi_j) / \text{sd}_M(\bar{Y}_j)$ for any non-degenerate mobility pattern. Both ratios concern variation across firms. Squaring the coefficient in the latter special case yields only the corresponding cross-firm variance ratio; it still does not identify the contribution of firm effects to individual-level healthcare-expenditure variation.

The empirical strategy follows directly from this distinction. I use mover event studies to assess pre-trends and document the timing and pass-through of healthcare expenditure after workers change firms. Firm-effect dispersion and its scale relative to individual-level expenditure variation are estimated directly from the limited-mobility-corrected two-way fixed-effects variance components.

A4 A Model

This appendix provides a simple Grossman-style health-capital model that micro-founds the log-linear two-way fixed-effects specification used in the main text. The aim is to show that, when health depreciation and the time cost of care-seeking have both individual and firm components, optimal medical spending can be written (in logs) as an additive function of an individual term, a firm term, and age. In this setting, the firm-specific component can be interpreted mainly as reflecting the workplace-specific health-depreciation rate, with a secondary role for differences in the time cost of accessing care.⁴⁸

I extend the life-cycle health demand framework of [Grossman \(1972b,a\)](#). Individuals do not derive direct utility from health; instead, health affects utility indirectly through time lost to sickness and through the resources (time and money) devoted to healthcare. This structure is well suited to the Dutch context, where basic health insurance is universal and largely independent of the employer. Firms affect health-related spending primarily by changing the health environment (workplace risks, stress, exposures) and, to a lesser extent, the time needed to obtain care, rather than the price or generosity of basic coverage (see Section II).

The model introduces two firm-specific elements. First, a firm-specific health-depreciation component, capturing how the work environment accelerates or slows health deterioration through physical risks, psychosocial stress, and environmental exposures. Second, a firm-specific time cost of care-seeking, reflecting workplace flexibility, ease of taking time off for appointments, and the presence of occupational health services or screening. In the Dutch institutional setting, the first channel (health depreciation) is likely to be the dominant component of firms' effects on basic-package spending, while the second channel is unlikely to generate large or systematic differences (see Section II). I nevertheless keep the time-cost term in the model to make clear that, in principle, firms can affect both the health-depreciation environment and the convenience of care-seeking; in the empirical analysis, I interpret the estimated firm effects mainly through the depreciation channel.

⁴⁸The model aims to explain how the workplace affects health and spending on healthcare, rather than why workers choose different firms.

Setup

Consider an employee working in firm j , with intertemporal utility

$$\sum_{t=0}^T \beta^t U(Z_t), \quad (6)$$

where T is the length of life,⁴⁹ β is the subjective discount factor, and Z_t is a composite of consumption goods and leisure:

$$Z_t = Z(C_t, L_t), \quad (7)$$

with C_t denoting market consumption and L_t leisure at age t .

Following Grossman (1972b), poor health reduces the available time endowment (normalised to 1). The employee faces the time constraint

$$T_t^w + T_t^h + L_t = 1 - T_t^{loss}, \quad (8)$$

$$T_t^{loss} = T^{loss}(H_t), \quad (9)$$

where H_t is the health stock at age t , T_t^{loss} the time lost due to sickness, and $T^{loss}(\cdot)$ a decreasing function of H_t . The variables T_t^w and T_t^h denote time spent working and time spent on health-care activities (care-seeking) at age t , respectively.

The health stock evolves according to

$$H_{t+1} - H_t = I_t - \delta_{t,j} H_t, \quad (10)$$

$$I_t = I(\Theta_t), \quad (11)$$

$$T_t^h = \chi_j \Theta_t, \quad (12)$$

where I_t is health investment at age t , $\delta_{t,j}$ is an age- and firm-specific depreciation rate, Θ_t is a composite of medical care goods, and χ_j measures the time cost of obtaining a given amount of medical care. A higher χ_j means more time must be spent on care-seeking per unit of Θ_t ; this reflects lower flexibility, more rigid

⁴⁹In line with Grossman (1972b,a), life continuation is predicated on maintaining a minimum health level, denoted as $\bar{H}$. Therefore, an individual's lifespan can be affected by their health and health investment decisions.

schedules, or less convenient access to care. In a firm with lower χ_j , employees can obtain the same care with less time loss.

At age zero (which may be interpreted as entry into the labour market), the individual faces the full-income budget constraint

$$\sum_{t=0}^T \frac{f_t \Theta_t + p_t C_t}{(1+r)^t} = \sum_{t=0}^T \frac{w_{t,j} T_t^w}{(1+r)^t} + A_0, \quad (13)$$

where f_t and p_t are the market prices of medical care and consumption goods, respectively, $w_{t,j} T_t^w$ is labour income at age t , $w_{t,j}$ is the wage rate in firm j , A_0 is initial wealth, and r is the interest rate. In the Dutch setting, basic-package prices and coverage are set nationally and do not vary across firms; I therefore treat f_t as homogeneous across firms, so that differences in medical spending attributable to firms arise through differences in quantities Θ_t rather than from price differences.

Normalising the discount factor so that $\beta = \frac{1}{1+r}$, the utility maximisation problem is equivalent to

$$\begin{aligned} & \max_{C_t, L_t, \Theta_t, T_t^h, T_t^w} \sum_{t=0}^T \frac{U(Z_t)}{(1+r)^t} \\ \text{s.t. } & \sum_{t=0}^T \frac{f_t \Theta_t + p_t C_t + w_{t,j} (T_t^h + L_t + T_t^{\text{loss}})}{(1+r)^t} = \underbrace{\sum_{t=0}^T \frac{w_{t,j}}{(1+r)^t} + A_0}_A, \end{aligned} \quad (14)$$

where the term $w_{t,j} (T_t^h + L_t + T_t^{\text{loss}})$ collects forgone earnings from time spent in care-seeking, leisure, and sickness. The right-hand side A represents full lifetime potential income (the value of all time at work plus initial wealth).

A4.I Optimal Health Investment Level

Focusing on health investments, consider the first-order condition with respect to $I_{\tau-1}$. In equilibrium,

$$\frac{\pi_{\tau-1}}{(1+r)^{\tau-1}} = - \sum_{t=\tau}^T w_{t,j} \frac{\frac{\partial H_t}{\partial I_{\tau-1}} \frac{dT_t^{\text{loss}}}{dH_t}(\cdot)}{(1+r)^t}, \quad (15)$$

where

$$\pi_\tau = f_\tau \frac{\partial \Theta_\tau}{\partial I_\tau} + w_{\tau,j} \frac{\partial T_\tau^h}{\partial I_\tau}$$

is the marginal cost of health investment (the combined monetary and time cost at age τ). Equation (15) states that the present value of the cost of an additional unit of health investment at age $\tau - 1$, $\pi_{\tau-1}$, equals the present value of the future gains from reduced time lost to sickness (and therefore higher effective labour supply and earnings).

Following the standard recursive manipulation in Grossman (1972b,a), this condition can be rewritten as

$$(1 + r)\pi_{\tau-1} = -w_{\tau,j} \frac{dT^{\text{loss}}}{dH_t}(\cdot) + (1 - \delta_{\tau,j})\pi_\tau, \quad (16)$$

or, equivalently,

$$\pi_{\tau-1} (r + \delta_{\tau,j} - \tilde{\pi}_\tau) = -w_{\tau,j} \frac{dT^{\text{loss}}}{dH_t}(\cdot), \quad (17)$$

where $\tilde{\pi}_\tau = \frac{\pi_\tau - \pi_{\tau-1}}{\pi_{\tau-1}}$ is the gross growth rate of the marginal cost of health.⁵⁰

A4.II Assumptions and a Closed-Form Demand Function

To obtain a closed-form expression for medical care demand, I introduce a set of standard simplifying assumptions.

(i) Linear investment in health goods. Assume health investments have constant returns to scale with respect to health goods:

$$I_\tau = I_0 \Theta_\tau, \quad (18)$$

⁵⁰For (17) to hold in this simple form, I follow Grossman (1972b,a) and assume that cross-terms of the form $\delta_{\tau,j} \tilde{\pi}_\tau$ are negligible. This keeps the focus on the main dependence on $\delta_{\tau,j}$ and $w_{\tau,j}$ without materially affecting the qualitative implications.

with $I_0 > 0$ a constant. Introducing diminishing returns would be more realistic but would prevent a simple closed-form solution. Under (18),

$$\pi_\tau = \frac{1}{I_0} (f_\tau + \chi_j w_{\tau,j}). \quad (19)$$

In the Dutch context, the monetary price component f_τ under the basic package is largely uniform across firms and, for high spenders, the marginal out-of-pocket price is often zero once the deductible is exhausted. For our purposes, any non-negligible but *firm-invariant* f_τ only affects the intercept of log spending and does not contribute to firm-level variation. To simplify the algebra and isolate firm-specific variation in care-seeking costs, I therefore treat the time-cost term $\chi_j w_{\tau,j}$ as the relevant driver of variation in π_τ across firms and write, up to a proportionality constant,

$$\pi_\tau \propto \chi_j w_{\tau,j}. \quad (20)$$

This implies that the growth rate of π_τ is approximately the growth rate of wages:

$$\tilde{\pi}_\tau \approx \tilde{w}_\tau. \quad (21)$$

(ii) Sick time as a function of health. Following the literature (e.g. [Galama and Van Kippersluis, 2019](#)), assume

$$T_\tau^{loss} = \phi H_\tau^{-\rho},$$

with $\phi > 0$ and $\rho > 0$. Substituting into (17) yields

$$\delta_{\tau,j} \frac{1}{I_0} \chi_j \left(1 + \frac{r - \tilde{w}_\tau}{\delta_{\tau,j}} \right) = \rho \phi H_\tau^{-(\rho+1)}. \quad (22)$$

(iii) Individual and firm components in depreciation and wages. Extend the functional forms commonly used in the life-cycle literature and assume

$$\delta_{\tau,j} = \delta_0 \delta_j e^{\kappa(\tau)},$$

where δ_0 and δ_j are individual- and firm-specific factors determining the level of health depreciation, and $\kappa(\tau)$ is an increasing function of age. By normalisation,

the population averages of δ_0 and δ_j equal one, so that $e^{\kappa(\tau)}$ is the average age profile of depreciation.

Similarly, following [Abowd et al. \(1999\)](#), assume wages factor as

$$w_{\tau,j} = w_0 w_j e^{l(\tau)},$$

where w_0 is a common level term, w_j a firm-specific wage component, and $l(\tau)$ an age profile. This implies $\tilde{w}_\tau \approx l'(\tau)$.

Taking logs and rearranging, we obtain

$$(1 + \rho) \log(H_\tau) \approx \log(\rho) + \log(\phi) + \log(I_0) - \log(\delta_0) - \log(\delta_j) - \log(\chi_j) - \kappa(\tau) - \log\left(1 + \frac{r - l'(\tau)}{\delta_{\tau,j}}\right). \quad (23)$$

Assuming either $(r - l'(\tau)) \ll \delta_{\tau,j}$ or, more generally, that the last term in (23) can be well approximated by a function of age only,⁵¹ we can write

$$\log(H_\tau) \approx \varepsilon \left[\log(\rho) + \log(\phi) + \log(I_0) - \log(\delta_0) - \log(\delta_j) - \log(\chi_j) - \check{\kappa}(\tau) \right], \quad (24)$$

where $\varepsilon = \frac{1}{\rho+1}$ and $\check{\kappa}(\tau)$ is an age function that absorbs $\kappa(\tau)$ and the term in brackets in (23).

Using (10) and (18), and approximating health around a local steady state so that $I_\tau \approx \delta_{\tau,j} H_\tau$, we obtain an expression for medical care goods:

$$\begin{aligned} \log(\Theta_\tau) \approx & \underbrace{\varepsilon \log(\rho) + \varepsilon \log(\phi)}_C + \underbrace{(\varepsilon - 1) \log(I_0) + (1 - \varepsilon) \log(\delta_0)}_{\alpha_0} \\ & + \underbrace{(1 - \varepsilon) \log(\delta_j) - \varepsilon \log(\chi_j)}_{\alpha_j} + \underbrace{\kappa(\tau) - \varepsilon \check{\kappa}(\tau)}_{\eta(\tau)} + \log\left(1 + \frac{\varepsilon \check{\kappa}'(\tau)}{\delta_{\tau,j}}\right). \end{aligned} \quad (25)$$

As before, assuming either $\varepsilon \check{\kappa}'(\tau) \ll \delta_{\tau,j}$ or that the last term in (25) can be

⁵¹Formally, one can either neglect the last term in (23) or absorb it into an age-specific term, provided that the dependence on δ_0 and δ_j is weak. This is a standard approximation in health-capital models and preserves the separability between age and individual/firm factors.

approximated by an age function,⁵² we obtain the compact expression

$$\log(\Theta_\tau) \approx C + \alpha_0 + \alpha_j + \check{\eta}(\tau), \quad (26)$$

where $\check{\eta}(\tau)$ is an age function and

$$\alpha_j = (1 - \varepsilon) \log(\delta_j) - \varepsilon \log(\chi_j).$$

In the Dutch basic-insurance system, the per-unit price of covered care is set nationally and does not vary across firms. Total basic-package spending at age t can therefore be written as $c_t \propto f_t \Theta_t$, with f_t common across firms. Taking logs and using (26), we obtain

$$\log c_t \approx \underbrace{\alpha_0 + C + \log f_t}_{\text{individual term}} + \underbrace{\alpha_j}_{\text{firm term}} + \underbrace{\check{\eta}(t)}_{\text{age term}}.$$

For a given worker i , let c_{it} denote basic-package healthcare costs at age t , and define $h_{it} = \log(c_{it} + 1)$ as in the main text. Relabelling the individual component as α_i , the firm component as ψ_j , and the age term as part of $x'_{it}\beta$ yields exactly the log-linear two-way fixed-effects structure used in the main text:

$$h_{it} = \log(c_{it} + 1) \approx \alpha_i + \psi_j + x'_{it}\beta + r_{it},$$

with r_{it} collecting idiosyncratic shocks.

Two main observations follow from this framework. First, in a Grossman-type health-investment model with firm-specific health depreciation and time costs of care-seeking, the logarithm of healthcare expenditures can be decomposed into an individual-specific term, a firm-specific term, and an age profile. The individual term reflects the individual-specific depreciation factor and returns to healthcare investments (through δ_0 and I_0). The firm-specific term α_j captures a scaled firm-specific depreciation factor δ_j and, in principle, the firm's flexibility for care-seeking through χ_j . In the Dutch institutional context, where basic prices and coverage are uniform across firms and time-related barriers to care are limited,

⁵²Here, I assume either the last term in equation (25) is negligible, or it can be absorbed into $\eta(\tau)$, leaving the dependence on δ_0 , δ_j , and χ_j unchanged.

δ_j provides the most natural interpretation of the estimated firm effects ψ_j in the main text as workplace-specific health-depreciation rates.

Second, although health and healthcare expenditures result from multiplicative interactions between individual and firm factors (for example through products of δ_0 , δ_j , and χ_j), taking logarithms maps these multiplicative effects into an additively separable function of individual and firm components plus an age term. This log-additive structure is precisely what underlies the AKM specification in the main analysis and justifies interpreting the firm fixed effects ψ_j as a common proportional shift in healthcare spending across workers within a firm.

A5 Leave-Out Variance Correction for the AKM Model

This appendix summarises how I implement the leave-out variance correction of [Kline et al. \(2020\)](#) (KSS) for the AKM specification in Equation (2). The goal is to obtain unbiased estimates of variance and covariance components of worker and firm effects in the presence of limited mobility.

A5.I Quadratic-form representation and plug-in bias

Write the AKM model in stacked form as

$$h = Zv + \varepsilon,$$

where h collects log healthcare expenditures h_{it} , Z is the design matrix containing worker and firm dummies and observed covariates, and v stacks all parameters, $v = (\alpha_1, \dots, \psi_1, \dots, \beta')'$. Let z'_o denote the o -th row of Z , and $S_{zz} = \sum_{o=1}^n z_o z'_o$ the Gram matrix.

Many objects of interest in the AKM decomposition can be written as quadratic forms

$$\theta = v' A v,$$

for some known symmetric matrix A ; for example, $\text{Var}(\psi_j)$ and $\text{Cov}(\alpha_i, \psi_j)$ correspond to particular choices of A .⁵³ The natural plug-in estimator is

$$\hat{\theta}_{PI} = \hat{v}' A \hat{v},$$

where $\hat{v}$ is the OLS estimator from Equation (2). As shown by [Kline et al. \(2020\)](#), $\hat{\theta}_{PI}$ is generally biased when worker mobility is limited: even if each element of $\hat{v}$ is unbiased, the quadratic form combines true dispersion and the dispersion of the estimation error.

Under weak conditions, KSS show that the bias of the plug-in estimator can be written as

$$\text{Bias}(\hat{\theta}_{PI}) = \mathbb{E}[\hat{\theta}_{PI}] - \theta = \sum_{o=1}^n z'_o S_{zz}^{-1} A S_{zz}^{-1} z_o \sigma_o^2,$$

⁵³See [Kline et al. \(2020\)](#) for explicit A matrices for standard AKM variance components.

where $\sigma_o^2 = \mathbb{E}[\varepsilon_o^2]$ is the observation-specific error variance. Limited mobility bias is thus a weighted sum of these variances, with weights given by the leverage-like terms $z_o' S_{zz}^{-1} A S_{zz}^{-1} z_o$.

A5.II Leave-out error-variance estimation

To obtain an unbiased estimator of θ , KSS propose estimating each σ_o^2 using a leave-one-out construction. Let $\hat{v}_{-o}$ denote the OLS estimator from Equation (2) computed on the sample that excludes observation o . The leave-out prediction for h_o is $z_o' \hat{v}_{-o}$ and the associated error-variance estimator is

$$\hat{\sigma}_o^2 = h_o(h_o - z_o' \hat{v}_{-o}). \quad (27)$$

By construction, h_o is independent of $\hat{v}_{-o}$ (because h_o is not used in its estimation), and $z_o' \hat{v}_{-o}$ is an unbiased predictor of h_o conditional on Z . It follows that $\mathbb{E}[\hat{\sigma}_o^2 | Z] = \sigma_o^2$, so (27) is an unbiased estimator of the observation-specific error variance; see [Kline et al. \(2020\)](#) for a formal derivation.

Given $\hat{\sigma}_o^2$, an unbiased estimator of θ is then

$$\hat{\theta}_{KSS} = \hat{v}' A \hat{v} - \sum_{o=1}^n z_o' S_{zz}^{-1} A S_{zz}^{-1} z_o \hat{\sigma}_o^2, \quad (28)$$

which satisfies $\mathbb{E}[\hat{\theta}_{KSS}] = \theta$ under the conditions in [Kline et al. \(2020\)](#). Intuitively, $\hat{\theta}_{KSS}$ subtracts an estimate of the plug-in bias from the naive quadratic form.

In large panels, recomputing $\hat{v}_{-o}$ separately for every o is infeasible. KSS show that for linear models the leave-one-out prediction error can be written as

$$h_o - z_o' \hat{v}_{-o} = \frac{h_o - z_o' \hat{v}}{1 - \ell_o},$$

where $\ell_o = z_o' S_{zz}^{-1} z_o$ is the leverage of observation o . This identity implies that the leave-out prediction $z_o' \hat{v}_{-o}$, and therefore $\hat{\sigma}_o^2$, can be computed from the full-sample OLS quantities without repeated re-estimation.

A5.III Implementation for the AKM decomposition

In practice, I implement the KSS correction as follows. First, I partial out the observed covariates x_{it} from the outcome by estimating Equation (2) and forming the residualised dependent variable $h_{it} - x'_{it}\hat{\beta}$. The dimensionality problem affects the worker and firm effects, not the coefficients on x_{it} , so $\hat{\beta}$ is estimated precisely.

Second, I apply the leave-out formulas of [Kline et al. \(2020\)](#) to the worker–firm match level, treating an entire match (rather than a single person–year observation) as the leave-out unit to allow for serial correlation of errors within matches, as recommended by KSS. Using the quadratic-form representation, I construct $\hat{\theta}_{KSS}$ in (28) for the variance of firm effects, $Var(\psi_j)$, and for the covariance between worker and firm effects, $Cov(\alpha_i, \psi_j)$.

Finally, because the AKM design matrix is very large, I use the random-projection approximation proposed by [Kline et al. \(2020\)](#), based on the database-friendly projections of [Achlioptas \(2003\)](#), to evaluate the quadratic forms in (28) efficiently. This approximation delivers estimates that are numerically very close to the exact leave-out quantities at a fraction of the computational cost; see [Kline et al. \(2020\)](#) for details and accuracy guarantees.

A6 Split-Sample Validation of Firm-Effect Dispersion

Estimated firm effects contain sampling error, and their precision varies with firm size and mobility. As an independent check on the KSS-corrected firm-effect variance, I estimate firm effects in two disjoint worker samples and compare the resulting estimates for firms observed in both samples. Following the independent-sample logic of [Kline \(2025\)](#), the covariance between the two estimates isolates their common firm component rather than split-specific estimation noise.

I randomly partition workers into two mutually exclusive sets, $S = 1$ and $S = 2$, and estimate the two-way fixed-effects model separately in each subsample, including all worker-years for workers in that subsample. I restrict attention to firms that belong to the intersection of the two split-sample connected sets. To ensure comparability across splits, I impose the same normalization in each estimation by recentering firm effects so that $\sum_i \hat{\psi}_{j(i,t)}^{(s)} = 0$ for $s \in \{1, 2\}$. This produces two firm-effect estimates for each firm in the common set,

$$\hat{\psi}_j^{(1)} = \psi_j + u_j^{(1)}, \quad \hat{\psi}_j^{(2)} = \psi_j + u_j^{(2)},$$

where $u_j^{(1)}$ and $u_j^{(2)}$ are estimation errors induced by sampling variation in the two disjoint worker sets. Under random partitioning and the maintained two-way fixed-effects assumptions, $u_j^{(1)} \perp u_j^{(2)} \mid \psi_j$ and $\mathbb{E}[u_j^{(s)} \mid \psi_j] = 0$ for $s \in \{1, 2\}$ ([Kline, 2025](#)).

The cross-split covariance therefore identifies the variance of latent firm effects on the common support:

$$\text{Cov}(\hat{\psi}_j^{(1)}, \hat{\psi}_j^{(2)}) = \text{Var}(\psi_j).$$

I use this identity to assess the full-sample KSS estimate. The cross-split covariance is 0.026, close to the KSS-corrected firm-effect variance in the main sample. The KSS estimates obtained separately in the two worker splits are also close to the full-sample estimate (Table [A17](#)).

Statistic	Split 1 (KSS)	Split 2 (KSS)
<i>Panel A. AKM variance components estimated within split (KSS-corrected)</i>		
Log Health Exp. Var.: $\text{var}(h_i)$	2.632 (Ref.)	2.622 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.030 (1.13 %)	0.029 (1.11 %)
Employee FE Var.: $\text{var}(\alpha_i)$	1.598 (60.73 %)	1.486 (56.68 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	−0.007	−0.004
Correlation $\rho(\alpha_i, \psi_j)$	−0.033	−0.020
Explained share: workers & firms	0.613	0.575
No. Movers	576,056	576,410
No. Firms	84,852	85,160
No. of Observations	8,023,188	8,021,807
Average Health Exp. : $\text{mean}(h_i)$	5.614	5.613
<i>Panel B. Cross-split covariance of firm effects (common firms)</i>		
$\text{cov}(\hat{\psi}_j^{(1)}, \hat{\psi}_j^{(2)})$	0.026	

Notes: Columns report leave-one-match-out KSS-corrected components (Kline et al., 2020) from the two-way fixed-effects model estimated separately in two random worker splits (all worker-years for workers in a split are included). The outcome is log annual healthcare expenditure; the sample covers full-time employees from 2009–2016. Shares in parentheses are relative to $\text{var}(h_i)$ within the corresponding split. Panel B reports the covariance between firm effects estimated in the two splits for firms observed in both. Under independent, mean-zero estimation error across splits, this covariance identifies the variance of the latent firm effects on the common support.

Table A17: Variance Decomposition of Log Healthcare Expenditure: Sample Splits

A7 LLM-based collar classification from education codes

Because the administrative records do not contain occupational titles, I proxy blue- versus white-collar status using educational specialization. The data include Dutch SOI education codes and, for each code, an education/program title in Dutch (OPLEIDINGSNAAM). I treat this education title as a concise summary of the type of work the individual is trained for, and I map it into a broad collar category.

I implement this mapping using an automated large-language-model (LLM) classification routine. For each education title, I query the OpenAI Chat Completions API (model `gpt-4o-mini`) with a fixed prompt that asks the model to assign the title to one of three categories: (1) *blue collar*, for programs that typically lead to manual, physical, trade, or craft occupations; (2) *white collar*, for programs that typically lead to office-based, professional, managerial, or administrative occupations; and (3) *borderline*, for ambiguous or mixed programs (for example, titles that plausibly combine technical and supervisory content or are too generic to map cleanly). The prompt restricts the model to return only the numeric label (1–3), with no accompanying text. I run the model with temperature zero and a short output constraint, so the classification is deterministic and machine-readable.

In the analysis, I collapse these labels into the collar proxy used in the subsample exercises. White-collar workers are those whose education titles are classified as category (2), and blue-collar workers are those classified as category (1). The borderline category (3) captures mixed or insufficiently specific education titles; they are excluded from collar subsample analysis.

This approach is motivated by the fact that the task is a coarse, low-dimensional coding problem: the goal is not to infer a specific occupation, but to distinguish whether a program is predominantly geared toward manual/trade work versus office/professional work. A growing literature finds that LLMs achieve human-comparable accuracy in such straightforward classification settings. [Gilardi et al. \(2023\)](#) show that GPT-4 outperforms human crowdworkers in text annotation. [Törnberg \(2023\)](#) documents expert-level reliability in political text coding. [Chang et al. \(2024\)](#) report similar evidence from large-scale evaluations in social science contexts. Since mapping education titles into broad collar categories is simpler than the nuanced political or attitudinal coding studied in these papers, the resulting

proxy is expected to be reliable for the purpose of stratifying the AKM estimates by worker type.

A8 Female Sample Analysis

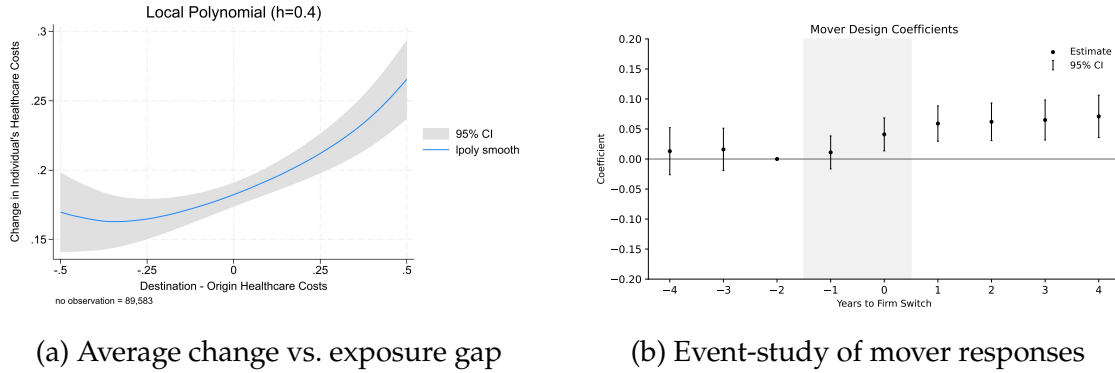


Figure A13: Workers' healthcare spending around moves between firms

Note: Sample of full-time female employees aged 25–65 who switch employer exactly once between 2009 and 2016 and belong to the connected set of firms. Healthcare outcomes are log total basic-package expenditures, residualised on age and calendar-year dummies. In panel (a), each point plots the average change in individual annuall residualised spending between the years before and the years after the move against the “exposure gap” D_i , defined as the difference in leave- i -out mean residualised spending between destination and origin firms; the line shows the (non-parametric) fitted relationship. In panel (b), the markers show coefficients from the mover event-study specification in equation (1), where the outcome is residualised log spending and the regressors are event-time indicators interacted with D_i , with event time measured relative to $t = -2$ as the reference year. Because workers are assigned to the firm where they spend most months, the year immediately before the move already contains some exposure to the destination firm.

Statistic	KSS Estimates
<i>Panel A. KSS variance components</i>	
Log Health Exp. Var.: $\text{var}(h_{it})$	2.617 (Ref.)
Firm FE Var.: $\text{var}(\psi_j)$	0.020 (0.78 %)
Employee FE Var.: $\text{var}(\alpha_i)$	3.158 (120.70 %)
Firm–Employee Cov.: $\text{cov}(\alpha_i, \psi_j)$	−0.004
Correlation $\rho(\alpha_i, \psi_j)$	−0.017
Explained share: workers & firms	1.211
No. Movers	366,643
No. Firms	44,578
No. of Observations	4,784,634
Average log Health Exp.: $\text{mean}(h_{it})$	6.114
<i>Panel B. Cross-sample link</i>	
Bias-corrected slope, firm FE (male) on firm FE (female) ^a	1.149
No. worker–year obs. in slope regression ^a	4,607,681

^a Slope from an OLS regression of the firm fixed effects estimated in the male subsample on the firm fixed effects estimated in the female subsample, using worker–year observations in the overlap where both firm effects are defined. The reported slope is bias-corrected by rescaling the attenuated OLS coefficient using the reliability ratio for the right-hand-side firm effect, $\text{Var}_{\text{KSS}}(\psi)/\text{Var}_{\text{PI}}(\hat{\psi})$, following [Kline et al. \(2020\)](#) (equivalently, $\hat{\beta}_{\text{BC}} = \hat{\beta}_{\text{OLS}} \times \text{Var}_{\text{PI}}(\hat{\psi}_{\text{female}})/\text{Var}_{\text{KSS}}(\psi_{\text{female}})$). In this application, $\hat{\beta}_{\text{OLS}} = 0.151$, $\text{Var}_{\text{PI}}(\hat{\psi}_{\text{female}}) = 0.155$, and $\text{Var}_{\text{KSS}}(\psi_{\text{female}}) = 0.020$, yielding $\hat{\beta}_{\text{BC}} = 1.149$. Because KSS is implemented on the leave-one-out connected set within each sex-specific subsample, the male and female samples need not form an exact partition of the baseline connected set; some firms appear in only one subsample and therefore drop out of the overlap regression.

Notes: Panel A components are estimated via leave-one-match-out KSS correction ([Kline et al., 2020](#)) using 50 random projections. The outcome is log annual basic-package health-care expenditure per calendar year, $h_{it} = \log(c_{it} + 1)$. Shares in parentheses are relative to $\text{var}(h_{it})$ and need not sum to 100% when the firm–worker covariance is negative.

Table A18: Variance Decomposition of Log Healthcare Expenditure (Females)